Faculty of Computing and AI, Air University, Islamabad

Department of Creative Technologies

Software Engineering

DevSync

Java Code Quality Analysis Platform

By

Ahsan Ilyas	221808/ISB
M. Abdul Raheem	221812/ISB
Ukasha Shazad	221818/ISB

Supervisor

Ms. Sabahat Ajaz

Bachelor of Science in Software Engineering (2022-2026)

The candidate confirms that the work submitted is their own and appropriate credit has been given where reference has been made to the work of others

Faculty of Computing and AI, Air University, Islamabad

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Software Engineering

DevSync

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of the requirement for the degree of

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By

Ahsan Ilyas	SP09-BSSE-221808/ISB
M. Abdul Raheem	SP09-BSSE-221812/ISB
Ukasha Shazad	SP09-BSSE-221818/ISB

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Ahsan Ilyas	M. Abdul Raheem	Ukasha Shazad

CERTIFICATE OF APPROVAL

It is to certify that the final year project of BS (SE) "DevSync - Java Code Quality Analysis Platform" was developed by **Ahsan Ilyas (221808/ISB)** and **M. Abdul Raheem (221812/ISB)** and **Ukasha Shazad (221818/ISB)** under the supervision of **"Ms. Sabahat Ajaz"** and that in their opinion; it is fully adequate, in scope and quality for the degree of Bachelors of Science in Software Engineering.

_____

Supervisor

_____

External Examiner

_____

Head of Department
(Department of Computer Science)

Executive Summary

DevSync is a comprehensive Java code quality analysis platform that addresses the critical need for automated code review and quality assessment in modern software development. The platform combines advanced Abstract Syntax Tree (AST) parsing with AI-powered recommendations to detect code smells, security vulnerabilities, and maintainability issues in Java projects.

Built with a modern React+Vite frontend and Spring Boot backend, DevSync serves as an automated code review assistant, enabling development teams to maintain high code quality standards throughout the software development lifecycle. The system implements 12+ sophisticated detectors for comprehensive code smell detection, including Long Method, Code Duplication, Broken Modularization, Complex Conditional, and Memory Leak detection and many more.

The platform features secure user authentication, file upload and processing capabilities supporting ZIP/JAR files up to 50MB, real-time analysis with progress tracking, AI-enhanced recommendations through Ollama integration, comprehensive report generation with visual diagrams, and historical analysis tracking for trend analysis.

DevSync provides a web-based dashboard with modern UI components, dark/light theme support, interactive data visualization, and responsive design optimized for all devices. The administrative interface enables user management, system monitoring, and configuration control.

The project successfully demonstrates the application of software engineering principles including Object-Oriented Programming, iterative development methodology, and comprehensive testing strategies. All 223 test cases passed successfully, confirming the system's reliability and functionality.

Acknowledgement

All praise is to Almighty Allah who bestowed upon us a minute portion of His boundless knowledge by virtue of which we were able to accomplish this challenging task.

We are greatly indebted to our project supervisor **Ms. Sabahat Ajaz**. Without their personal supervision, advice and valuable guidance, completion of this project would have been doubtful. We are grateful to them for their encouragement and continual help during this work.

And we are also thankful to our parents and family who have been a constant source of encouragement for us and brought us with the values of honesty & hard work.

Ahsan Ilyas	M. Abdul Raheem	Ukasha Shazad

Abbreviations

Abbreviation	Description
AST	Abstract Syntax Tree
API	Application Programming Interface
AI	Artificial Intelligence
CRUD	Create, Read, Update, Delete
CSS	Cascading Style Sheets
CORS	Cross-Origin Resource Sharing
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
IDE	Integrated Development Environment
JAR	Java Archive
JDK	Java Development Kit
JPA	Java Persistence API
JSON	JavaScript Object Notation
JWT	JSON Web Token
KLOC	Thousand Lines of Code
LOC	Lines of Code
MVC	Model-View-Controller
OOP	Object-Oriented Programming
ORM	Object-Relational Mapping
PDF	Portable Document Format
REST	Representational State Transfer
SQL	Structured Query Language
UI	User Interface
URL	Uniform Resource Locator
ZIP	Zone Improvement Plan (file compression format)

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1 Introduction

DevSync is a comprehensive Java code quality analysis platform that combines advanced Abstract Syntax Tree (AST) parsing with AI-powered recommendations to detect code smells, security vulnerabilities, and maintainability issues in Java projects. Built with a modern React+Vite frontend and Spring Boot backend, the platform serves as an automated code review assistant, enabling development teams to maintain high code quality standards throughout the software development lifecycle.

1.1 Project Scope

The project encompasses multiple interconnected modules designed to provide end-to-end code quality assessment capabilities.

1.1.1 Completed Modules

User Authentication System Secure user registration, login, and session management with BCrypt password encryption, ensuring data protection and user privacy compliance. Integrated with React Router DOM for seamless navigation and protected routes.

File Upload and Processing ZIP/JAR file upload with drag-and-drop support, extraction, and Java file collection capabilities supporting project structures up to 50MB. Real-time upload progress tracking with visual feedback.

Advanced Code Analysis Engine Twelve sophisticated detectors implementing complex algorithms for comprehensive code smell detection:

- Long Method Detector (cyclomatic and cognitive complexity)
- Code Duplication Detector
- Broken Modularization Detector (cohesion analysis)
- Complex Conditional Detector
- Deficient Encapsulation Detector
- Empty Catch Block Detector
- Long Identifier Detector
- Long Parameter List Detector
- Long Statement Detector
- Magic Number Detector
- Missing Default Case Detector
- Unnecessary Abstraction Detector

AI Integration Module Ollama AI service integration with fallback mechanisms for intelligent code recommendations. Provides context-aware suggestions for refactoring and code improvement when external services are unavailable.

Report Generation System Comprehensive text-based reporting with severity categorization (Critical, Warning, Suggestion), visual diagram generation using PlantUML, and PDF export capabilities using jsPDF. Reports include executive summaries, detailed issue breakdowns, and actionable recommendations.

Analysis History Tracking Persistent storage and retrieval of analysis sessions with detailed metrics, enabling trend analysis, code quality improvement tracking, and comparative analysis across multiple project versions.

Web-based Dashboard Modern React 18.3.1 + Vite 7.1.7 frontend featuring:

- Radix UI component library for accessible, production-ready components
- Tailwind CSS v4 for utility-first styling with PostCSS optimization
- next-themes for seamless dark/light mode switching
- Recharts for interactive data visualization and analytics
- Framer Motion for smooth animations and transitions
- React Hook Form for efficient form validation
- Lucide React icons for consistent iconography
- Responsive design optimized for desktop, tablet, and mobile devices

Admin Panel Administrative interface for user management, system monitoring, configuration control, and analytics dashboard with real-time metrics visualization.

1.1.2 Key Features Implemented

The platform incorporates several advanced features that distinguish it from conventional static analysis tools:

Multi-dimensional Code Quality Assessment Utilizes both cyclomatic and cognitive complexity metrics to provide nuanced code quality evaluation beyond simple line counts.

Real-time AST Parsing Integration with JavaParser library enables immediate Abstract Syntax Tree generation and analysis upon file upload, with progress tracking and visual feedback in the UI.

Severity-based Issue Classification Three-tier classification system (Critical, Warning, Suggestion) prioritizes issues based on their impact on code quality and maintainability.

Historical Analysis Tracking Comparison capabilities allow developers to track code quality improvements over time and identify regression patterns.

AI-enhanced Recommendations Contextual suggestions powered by Ollama AI provide actionable guidance for addressing identified issues, with intelligent fallback mechanisms ensuring continuous service availability.

Secure File Handling Access control mechanisms with JWT-based authentication ensure that uploaded code remains protected and accessible only to authorized users. Files are stored with unique identifiers and proper access validation.

2 Design Methodology and Software Process Model

2.1 Design Methodology: Object-Oriented Programming

The DevSync project follows Object-Oriented Programming (OOP) principles, providing a robust foundation for building maintainable and extensible software. This methodology was selected based on the following justifications:

2.1.1 Encapsulation

Each detector class, such as `LongMethodDetector` and `CodeDuplicationDetector`, encapsulates specific analysis logic and maintains internal state through private methods and fields. This design principle ensures data integrity by preventing unauthorized access to internal implementation details while providing clear, well-defined interfaces for module interaction. For example, the complexity calculation algorithms remain hidden within detector classes, exposing only the final analysis results through public methods.

2.1.2 Inheritance

The project utilizes inheritance through Spring Boot's component hierarchy and JPA entity relationships. All detectors share common analysis patterns inherited from base classes while extending functionality for specific detection scenarios. This hierarchical structure reduces code duplication and ensures consistent behavior across the detector family.

2.1.3 Polymorphism

The detector system demonstrates polymorphism where different detector implementations are processed uniformly through common interfaces. This enables flexible execution of the analysis pipeline, allowing new detectors to be added without modifying existing orchestration logic. The runtime binding of detector implementations supports extensibility and testing through mock implementations.

2.1.4 Abstraction

Complex analysis algorithms are abstracted into high-level interfaces, hiding implementation details while providing clear contracts for functionality. Users of the analysis engine interact with simplified APIs without needing to understand the underlying AST traversal algorithms or metric calculation formulas.

2.2 Software Process Model: Iterative and Incremental Development

The project follows an Iterative and Incremental Development model, chosen for its flexibility and risk mitigation capabilities in complex software projects.

2.2.1 Iterative Approach

The system was developed through multiple iterations, with each iteration focusing on specific functionality:

1. **Iteration 1:** Core authentication and user management
2. **Iteration 2:** File processing and extraction pipeline
3. **Iteration 3:** Basic code analysis with initial detectors
4. **Iteration 4:** Report generation and visualization
5. **Iteration 5:** AI integration and advanced recommendations
6. **Iteration 6:** Admin panel and system monitoring

This approach allowed continuous refinement and improvement of existing features based on feedback from each iteration.

2.2.2 Incremental Delivery

Features were delivered incrementally, starting with core functionality and progressively adding advanced capabilities. This enabled early testing and validation, ensuring that foundational components were stable before building dependent features.

2.2.3 Risk Mitigation

The iterative approach allowed early identification and resolution of technical challenges, particularly in AST parsing complexity and AI service integration. Prototyping during early iterations exposed potential issues before significant development investment.

2.2.4 Flexibility

The incremental model provided flexibility to adapt requirements based on testing feedback and emerging needs. Fallback mechanisms for AI service unavailability were added in response to reliability concerns identified during integration testing.

3 System Overview

DevSync is an intelligent Java code analysis platform designed to improve software quality through automated detection of code smells, security vulnerabilities, and maintainability issues. The system combines traditional static analysis techniques with modern AI-powered insights to provide comprehensive code quality assessment.

3.1 Functionality Context

The platform analyzes uploaded Java project files and generates detailed reports containing actionable recommendations. Users can track code quality improvements over time through historical analysis comparison and receive AI-generated suggestions for code enhancement. The workflow supports individual developers performing personal code reviews as well as team leads monitoring project-wide code quality metrics.

3.2 Design Context

The system architecture follows a client-server model where a React.js frontend communicates with a Spring Boot backend through RESTful APIs. The backend integrates multiple analysis engines and external AI services, providing a unified interface for code quality assessment regardless of the underlying analysis mechanism.

3.3 Background Information

Modern software development faces increasing complexity in maintaining code quality across large codebases. Manual code reviews are time-consuming and may miss subtle issues that automated tools can detect consistently. DevSync addresses this challenge by automating code analysis using advanced algorithms and AI-driven recommendations, reducing the burden on human reviewers while improving detection coverage.

3.4 Architectural Design

3.4.1 Modular Program Structure

The DevSync system is decomposed into four major modules, each responsible for distinct aspects of the platform's functionality.

Frontend Module (React+Vite) The presentation layer consists of React components organized by functionality, built with modern tooling:

Authentication Components Handle user login, registration, and session management with secure token storage using React Router DOM for navigation.

Dashboard Components Display analysis results with interactive visualizations using Recharts, historical trends, and quick-access functionality with smooth transitions powered by Framer Motion.

Reusable UI Components Comprehensive component library built with Radix UI primitives including dialogs, dropdowns, accordions, tabs, tooltips, and more. Styled with Tailwind CSS for consistent design and full accessibility compliance.

Theme System Dynamic dark/light mode switching using next-themes with persistent user preferences and seamless transitions.

API Integration Layer Manages all backend communication using Axios (v1.12.2) with request interceptors, error handling, and response transformation.

Build System Vite-powered development environment providing instant hot module replacement (HMR), optimized production builds, and fast development server.

Backend Module (Spring Boot) The application layer implements business logic through a layered architecture:

Controller Layer REST endpoints handling HTTP requests with validation and response formatting.

Service Layer Business logic implementation with transaction management and cross-cutting concerns.

Repository Layer Data access abstraction using Spring Data JPA with custom query methods.

Analysis Engine Orchestrates detector execution and result aggregation.

Analysis Engine Module The core processing component responsible for code quality assessment:

AST Parser JavaParser integration for source code parsing and syntax tree generation.

Detector Components Specialized analyzers for different code smell categories.

Report Generator Formats analysis results into human-readable reports with visual diagrams.

AI Integration Ollama service connection with graceful degradation for service unavailability.

Data Management Module Persistence and storage management:

User Management Account data, preferences, and access control.

Analysis History Tracking of past analyses with comparison capabilities.

File Management Secure storage and retrieval of uploaded projects and generated reports.

3.4.2 Module Relationships and Collaboration

The system modules collaborate through well-defined interfaces to provide comprehensive code analysis functionality. The frontend communicates with the backend through RESTful APIs for authentication, file uploads, and result retrieval. Controllers delegate business logic to services, which interact with repositories for data persistence. The analysis engine coordinates detector execution, report generation, and AI service communication with fallback mechanisms.

Data flows unidirectionally from user input through processing layers to storage, with query paths returning results through the same layered structure. This separation ensures that changes to one layer do not cascade unnecessarily to others.

Note: The detailed box-and-line diagram is provided in Appendix A.

3.4.3 Detailed Architecture Mapping

The system implements a four-tier architecture with clear responsibility boundaries:

Client Tier (Presentation Layer)

React Components Functional components with hooks for state and lifecycle management, utilizing React 18.3.1 features.

State Management Context API for global state with local component state for UI interactions, enhanced with React Hook Form for form validation.

API Client Centralized Axios instance with interceptors for authentication and error handling.

UI Component Library Radix UI primitives providing accessible, unstyled components including Accordion, Alert Dialog, Avatar, Checkbox, Dialog, Dropdown Menu, Navigation Menu, Popover, Progress, Radio Group, Scroll Area, Select, Slider, Switch, Tabs, Toggle, and Tooltip.

Styling System Tailwind CSS v4.1.16 with PostCSS for utility-first styling, class-variance-authority for component variants, and tailwind-merge for class name optimization.

Theme System next-themes library providing seamless dark/light mode switching with system preference detection and persistent storage.

Visualization Recharts library for interactive data visualization and analytics dashboards.

Animation Framer Motion for smooth, performant animations and transitions throughout the interface.

Server Tier (Application Layer)

Web Layer Spring MVC controllers with request validation and response serialization.

Business Layer Service classes implementing domain logic with transactional boundaries.

Integration Layer External service connectors for AI and potential future integrations.

Security Layer Spring Security configuration with JWT-based authentication.

Data Tier (Persistence Layer)

Repository Layer JPA repositories with custom queries for complex data retrieval.

Entity Layer Domain entities with JPA annotations and validation constraints.

Database Layer MySQL database with optimized schema design.

File Storage File system storage for uploaded projects and generated reports.

Analysis Tier (Processing Layer)

Parser Layer JavaParser integration for AST generation from source files.

Detection Layer Individual detector implementations for specific code smells.

Analysis Layer Orchestration of detector execution and result aggregation.

Reporting Layer Report formatting and diagram generation.

This layered architecture ensures separation of concerns, maintainability, and scalability while providing clear interfaces between system components.

4 Design Models

4.1 Activity Diagram: User Registration

This activity diagram explains how a user interacts with the system during registration and login. First, the user provides basic credentials. The system then checks whether the user already exists in the database. If the user is new, the registration form is displayed and the entered data goes through a validation process to ensure correctness. After successful validation, the user is

granted access to the dashboard. In case of invalid data, the system stops the process and shows an error message.

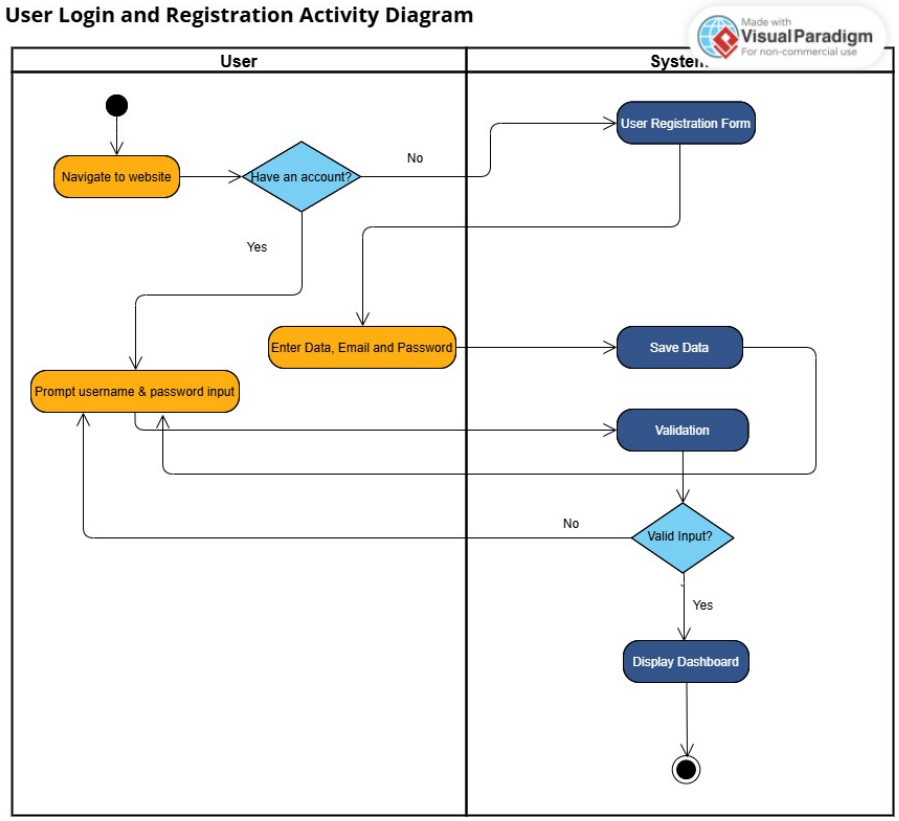

Figure 1: Activity Diagram for User Registration and Login

4.2 Activity Diagram: Code Analysis

This diagram represents the main functionality of the DevSync system. The process starts when a developer uploads a ZIP file containing source code. The system extracts the files and runs multiple detectors to identify code issues such as smells or bad practices. These detectors work simultaneously to save time. After analysis, the system communicates with the local AI service (Ollama) to generate improvement suggestions. If the AI service is unavailable, the system follows a fallback mechanism to ensure that a report is still generated.

4.3 Class Diagram: Whole System

The class diagram provides a structural overview of the DevSync system. It shows the main classes, their attributes, and methods, along with the relationships between them. Key components such as the UserController, UploadController, and ReportGenerator coordinate system behavior. The diagram also highlights how data is managed and stored in the database. Overall, this diagram acts as a blueprint that helps developers understand how different parts of the system are organized.

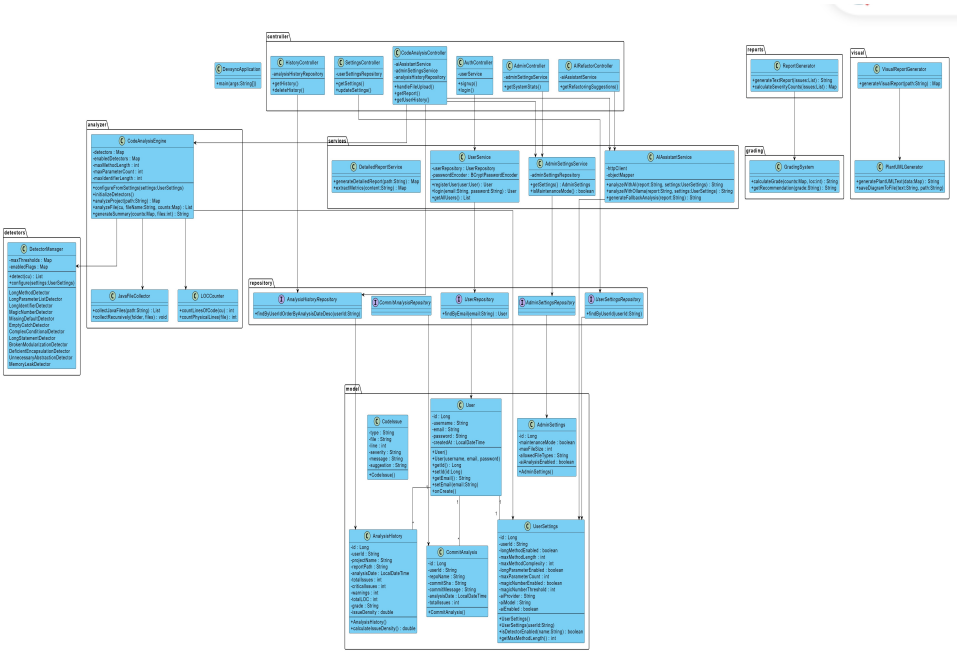

Figure 3: Class Diagram for DevSync System

4.4 Sequence Diagram: User Registration

This sequence diagram illustrates the step-by-step interaction involved in user registration and login. It shows how the user communicates with the graphical interface, which then forwards requests to the authentication controller. The controller validates the data by interacting with the database. An alternative (Alt) flow is included to handle both successful and failed login scenarios, clearly showing system behavior in each case.

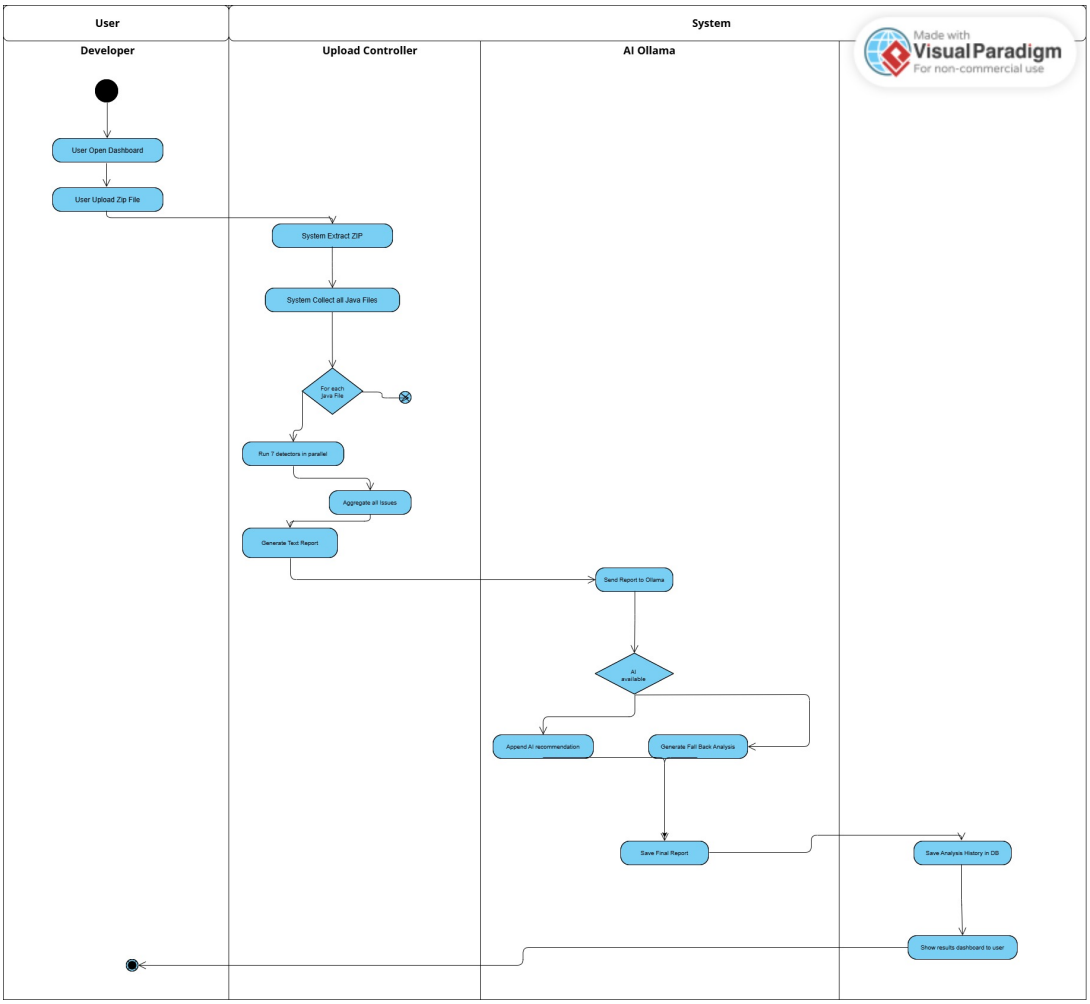

Figure 2: Activity Diagram for Code Analysis Workflow

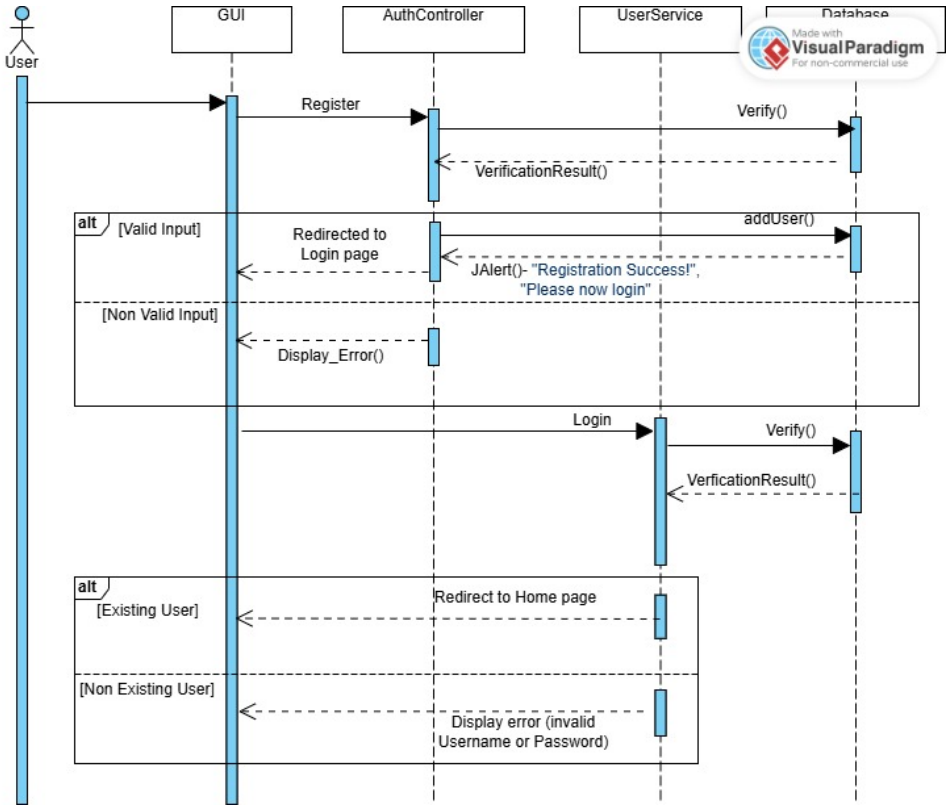

Figure 4: Sequence Diagram for User Registration

4.5 Sequence Diagram: Code Analysis and AI Feedback

This sequence diagram focuses on the detailed workflow of code analysis. The UploadController manages the entire process by coordinating file extraction, detector execution, and communication with the AI service. Once analysis is complete, the generated report is saved in the system history and then displayed to the user. This diagram highlights how different components interact in a well-defined order to complete the analysis task.

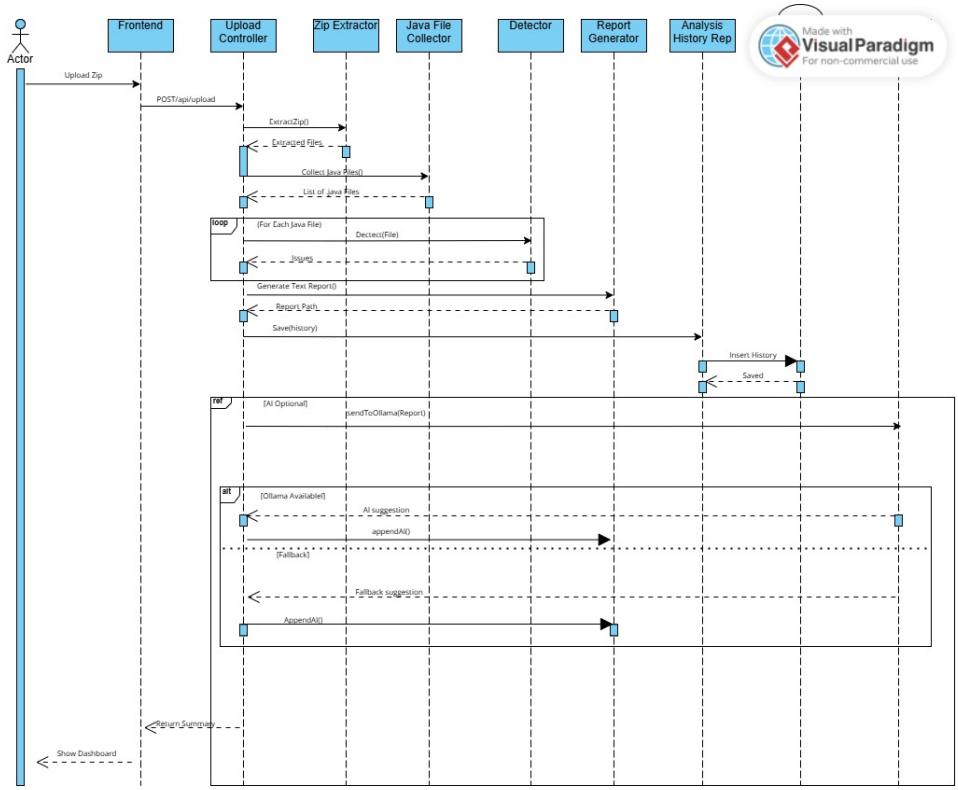

Figure 5: Detailed Sequence for Code Analysis and AI Integration

4.6 System Sequence Diagram (SSD)

The system sequence diagram presents a high-level view of the system from an external perspective. It treats the system as a black box and only shows interactions between the developer and the system. The diagram covers major events such as uploading source code, starting the analysis process, and retrieving past analysis results from history.

Figure 6: System Sequence Diagram (High Level)

4.7 State Transition Diagram: System States

This diagram describes how the system changes its state during execution. Initially, the system remains in an idle state. When a file is uploaded, it moves to the uploading state, followed by the analyzing state. The diagram also includes error states, such as invalid files or detector failures, and shows how the system either recovers from these errors or safely returns to the

idle state.

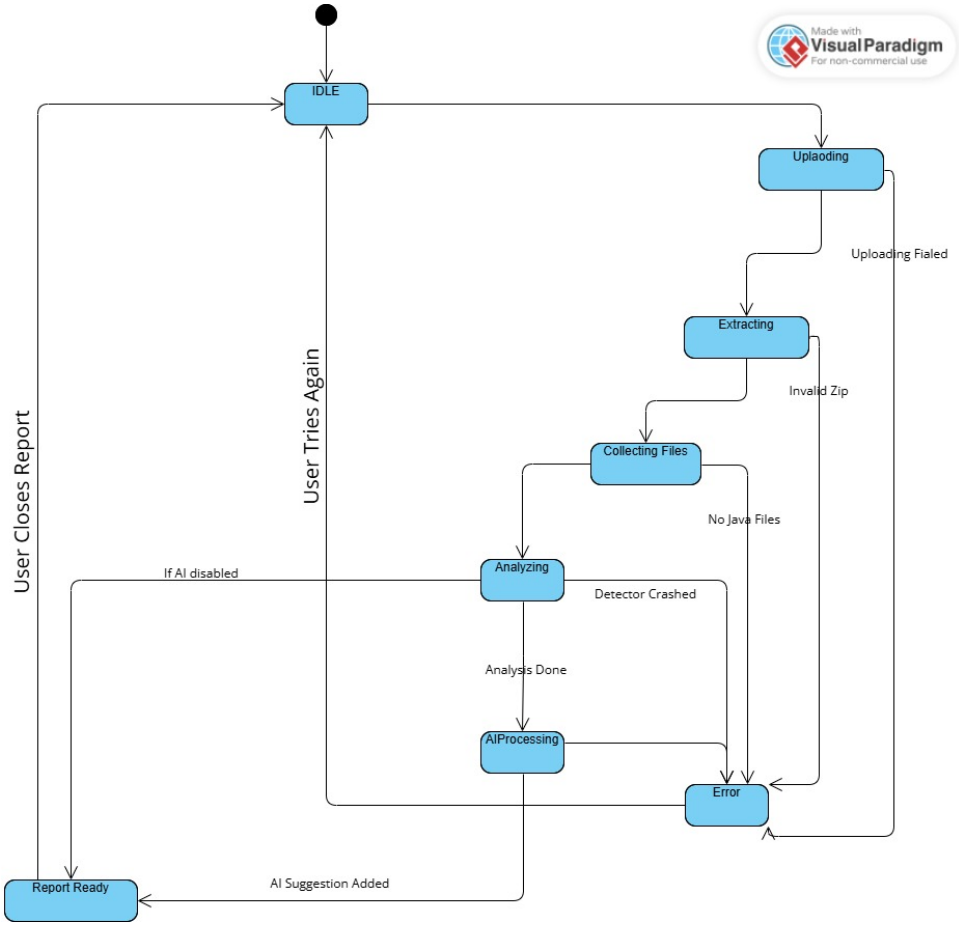

Figure 7: State Transition Diagram of the System

5 Data Design

5.1 Overview

DevSync is a code quality analysis system that transforms uploaded Java projects into structured analysis reports. The system processes source code files, detects code smells using twelve specialized detectors, and stores analysis results in a relational database while maintaining file-based report storage. This hybrid approach balances query performance with storage efficiency for large report files.

5.2 Data Architecture

5.2.1 High-Level Data Flow

The data transformation pipeline processes user uploads through multiple stages, each refining the data representation:

```
User Upload (ZIP/JAR) via React Frontend
        |
        v
Vite Dev Server / Production Build
        |
        v
Axios HTTP Request to Spring Boot Backend
        |
        v
File Extraction & Validation
        |
        v
Java Source Code Parsing (JavaParser AST)
        |
        v
Code Analysis Engine (12 Detectors)
        |
        v
Issue Detection & Scoring
        |
        v
Ollama AI Recommendations (with fallback)
        |
        v
Report Generation (TXT/Visual/PDF)
        |
        v
Database Persistence + File Storage
        |
        v
JSON Response to Frontend
        |
        v
Recharts Visualization & UI Presentation
```

5.2.2 System Layers

The data architecture spans six distinct layers, each with specific responsibilities:

1. **Presentation Layer:** React 18.3.1 + Vite 7.1.7 frontend with Radix UI components, Tailwind CSS styling, and Recharts visualization. Handles user interaction, data visualization, and responsive design across devices.

2. **Communication Layer:** Axios HTTP client with interceptors for authentication, error handling, and request/response transformation. Manages REST API communication between frontend and backend.

3. **Application Layer:** Spring Boot controllers and services manage request routing, business orchestration, and transaction boundaries.

4. **Business Logic Layer:** Code analysis engine with twelve specialized detectors implements core analysis algorithms, AST traversal, and metric calculation.

5. **Data Access Layer:** JPA repositories provide abstracted database operations with transaction management and query optimization.

6. **Persistence Layer:** MySQL database and file system storage maintain durable data records with proper indexing and access control.

5.3 Data Transformation and Processing

5.3.1 Input Data Transformation

The system processes input data through six sequential stages, each transforming the data into increasingly refined representations.

Stage 1: File Upload Processing **Input:** Multipart file (ZIP/JAR format, maximum 50MB) uploaded via React frontend with drag-and-drop support or file picker

Process:

- Frontend validates file type and size before upload
- Axios sends multipart/form-data request with progress tracking
- Backend extracts compressed files to a temporary directory
- Generate a unique folder name using timestamp and UUID
- Validate project file structure for expected Java project layout
- Verify file integrity and scan for malformed archives
- Real-time progress updates sent to frontend via WebSocket or polling

Output: Extracted project directory structure with validated contents and upload confirmation

Stage 2: Source Code Parsing **Input:** Java source files (.java extension)
Process:

- JavaParser generates Abstract Syntax Tree (AST) for each source file
- CompilationUnit objects created containing parsed syntax structure
- AST nodes extracted including classes, methods, fields, and statements
- Symbol resolution performed for type reference analysis

Output: Structured CompilationUnit objects representing source code

Stage 3: Code Analysis **Input:** CompilationUnit AST objects
Process:

- Twelve detector instances analyze AST in parallel where possible
- Detector-specific algorithms traverse relevant AST nodes
- Metrics calculated including LOC, complexity, cohesion, and coupling
- Threshold comparison identifies violations and generates issues

Output: Collection of CodeIssue objects with location and severity

Stage 4: Issue Aggregation **Input:** Raw issue strings from detector components
Process:

- Parse issue format: `[Severity] [Type] File:Line - Message`
- Extract structured fields into typed objects
- Calculate severity distribution counts
- Compute grading metrics based on issue weights

Output: Structured analysis results map with aggregated statistics

Stage 5: Report Generation **Input:** Analysis results map with issue data and AI recommendations
Process:

- Format issues into human-readable text with contextual information
- Generate PlantUML class and sequence diagrams
- Calculate summary statistics and overall quality grades

- Create comprehensive report with executive summary
- Generate PDF version using jsPDF for download capability
- Prepare JSON response with structured data for frontend visualization

Output: Text report file, visual diagram images, PDF document, and JSON data for Recharts visualization

Stage 6: Data Persistence and Presentation **Input:** Analysis results, metadata, and generated reports

Process:

- Map results to JPA entity objects
- Validate constraints and referential integrity
- Execute database transactions with rollback support
- Store report files to designated file system location
- Send JSON response to frontend via Axios
- Frontend processes data for Recharts visualization
- Update UI with analysis results, charts, and downloadable reports
- Apply theme-aware styling using next-themes
- Animate transitions using Framer Motion

Output: Persisted database records, stored report files, and interactive dashboard with visualizations

5.3.2 Data Processing Algorithms

The analysis engine employs several key algorithms for code quality assessment:

Cohesion Index Calculation

```
cohesionIndex = usedFields / totalFields
```

This metric measures how comprehensively methods utilize class fields. Values range from 0.0 (low cohesion, indicating potential god class) to 1.0 (high cohesion, indicating focused responsibility). The system flags classes with cohesion index below 0.4 as exhibiting broken modularization.

Coupling Count Calculation

```
couplingCount = count(unique external dependencies)
```

This metric counts distinct external class references within a class. High coupling (values greater than 6) indicates excessive dependencies that reduce maintainability and increase change impact.

Cyclomatic Complexity

```
complexity = 1 + count(decision_points)
```

Cyclomatic complexity measures the number of independent execution paths through a method. Decision points include if statements, loops, case labels, and boolean operators. Methods with complexity above 10 are flagged as requiring refactoring.

Cognitive Complexity

```
cognitiveComplexity = sum(nesting_penalties) + sum(structural_penalties)
```

Cognitive complexity measures human comprehension difficulty, accounting for nesting depth and control flow breaks. Unlike cyclomatic complexity, it penalizes deeply nested structures more heavily. Values above 15 indicate hard-to-understand code requiring simplification.

Issue Density

```
issueDensity = totalIssues / (totalLOC / 1000)
```

Issue density normalizes the issue count per thousand lines of code, enabling fair comparison across projects of different sizes.

Grading Algorithm

```
score = 100 - (critical * 10 + warning * 5 + suggestion * 2)
```

The grading system assigns letter grades based on weighted issue counts:

- **A+ (95–100):** Excellent code quality with minimal issues
- **A (90–94):** Very good quality with few minor issues
- **B (80–89):** Good quality with some improvement areas

- **C (70–79):** Acceptable quality requiring attention
- **D (60–69):** Below average quality with significant issues
- **F (below 60):** Poor quality requiring major refactoring

5.4 Data Storage

5.4.1 Database Storage (MySQL)

The system uses MySQL as the primary relational database with the following configuration:

Database Name: `devsyncdb`

The database stores persistent system data including users, settings, and analysis history. Core tables include:

users User account information including credentials and profile data

user_settings Per-user configuration preferences and notification settings

analysis_history Records of all analysis sessions with summary metrics

commit_analysis Git commit-level analysis results for repository tracking

admin_settings System-wide configuration parameters

5.4.2 File System Storage

Report files and uploaded projects are stored in a structured file system hierarchy:

Base Directory: `uploads/`

```
uploads/
  +-- {timestamp}/
  |     +-- {project_name}/
  |     +-- report.txt
  |     +-- diagrams/
  |           +-- class_diagram.png
  |           +-- sequence_diagram.png
```

Files are organized using timestamp-based unique folders to prevent naming collisions. Report file paths are stored in the database and served through backend controllers with appropriate access control.

5.4.3 In-Memory Processing

During analysis, temporary data structures are utilized for processing efficiency:

- Analysis results map (cleared after report generation to free memory)
- Singleton detector instances (reused across analyses for efficiency)
- AST objects retained only during the analysis lifecycle
- Caching of frequently accessed configuration values

5.5 Summary

The DevSync data design implements a robust and scalable architecture that transforms uploaded source code into structured AST representations, processes them through twelve configurable detectors, and stores results using a hybrid database and file-based approach. The modern React 18.3.1 + Vite 7.1.7 frontend provides an intuitive, accessible interface with:

- Radix UI component library for production-ready, accessible components
- Tailwind CSS v4 for efficient, utility-first styling
- next-themes for seamless dark/light mode switching
- Recharts for interactive data visualization
- Framer Motion for smooth animations and transitions
- jsPDF for client-side PDF generation
- React Hook Form for efficient form validation

The design supports extensibility through modular detector architecture, efficient processing through parallel analysis where applicable, and long-term analysis tracking through persistent storage. The separation of transactional data (database) from large artifacts (file system) optimizes both query performance and storage efficiency while maintaining system maintainability. The frontend-backend separation enables independent scaling, technology updates, and enhanced user experience through modern web technologies.

6 Human Interface Design

6.1 System Functionality from User Perspective

6.1.1 Landing Page Experience

Users are greeted with a comprehensive landing page that showcases DevSync's capabilities through:

Hero Section Prominent call-to-action with clear value proposition - "Eliminate Java Code Smells" with gradient text effects and animated background elements that create visual engagement without distraction.

Feature Showcase Interactive sections highlighting key capabilities:

- AST-based code analysis with visual code examples
- AI-powered recommendations with mock analysis results
- Multi-dimensional quality assessment with metric explanations
- Historical tracking capabilities with trend visualization

How-to-Use Guide Step-by-step process visualization:

1. Sign up and login with visual indicators
2. Upload project with drag-and-drop demonstration
3. Receive analysis with sample report preview
4. Collaborate with team sharing features

Theme Switching Users can toggle between dark and light modes with smooth transitions, ensuring comfortable viewing in different environments and personal preferences.

6.1.2 Authentication Flow

Registration Process Streamlined signup with real-time validation:

- Username availability checking with immediate feedback
- Email format validation with clear error messaging
- Password strength indicators with security requirements
- Success confirmation with automatic redirection

Login Interface Secure authentication with user-friendly features:

- Email/password combination with remember me option
- Clear error messaging for invalid credentials
- Social login placeholders for future OAuth integration
- Password recovery options (planned enhancement)

6.1.3 Main Dashboard Experience

Upload Interface Intuitive file upload with multiple interaction methods:

- Drag-and-drop zone with visual feedback during hover and drop states
- Traditional file browser integration with file type filtering
- Upload progress indication with real-time status updates
- File validation with clear acceptance criteria (ZIP files only)

Analysis Progress Real-time feedback during processing:

- Progress indicators showing current analysis stage
- Estimated completion time based on project size
- Cancellation option for long-running analyses
- Error handling with detailed failure explanations

Results Visualization Comprehensive analysis presentation:

- Severity-based issue categorization with color coding (Critical, Warning, Suggestion)
- Numerical summaries with visual progress bars
- Detailed analysis summary in formatted text blocks
- Interactive report viewing with modal dialogs

6.1.4 Historical Analysis Management

History Panel Chronological analysis tracking:

- Timeline view of previous analyses with project names and dates
- Quick metrics overview for each analysis session
- Filtering and search capabilities for large analysis histories
- Comparison tools for tracking improvement trends

Report Access Seamless report retrieval and viewing:

- One-click access to detailed analysis reports
- Full-screen report viewing with syntax highlighting
- Download options for offline review and sharing
- Print-friendly formatting for documentation purposes

6.1.5 Administrative Interface

Admin Dashboard Comprehensive system management:

- User account overview with registration statistics
- System performance metrics with real-time monitoring
- Analysis volume tracking with usage patterns
- Administrative controls for user management

User Management Detailed user administration:

- User account listing with search and filter capabilities
- Account status management with activation/deactivation controls
- Usage statistics per user with analysis frequency tracking
- Security monitoring with login attempt tracking

6.2 Feedback Information Display

6.2.1 Success Feedback

Analysis Completion Clear success indicators with actionable next steps:

- Completion notifications with issue count summaries
- Success animations with visual confirmation
- Immediate access to results with prominent view buttons
- Sharing options for team collaboration

User Actions Positive reinforcement for user interactions:

- Registration success with welcome messaging
- Login confirmation with personalized greetings
- File upload success with processing initiation
- Settings updates with immediate effect confirmation

6.2.2 Error Handling and Guidance

Validation Errors Helpful error messaging with correction guidance:

- Form validation with field-specific error indicators
- File upload errors with format and size requirement explanations
- Authentication failures with clear resolution steps
- Network errors with retry mechanisms and offline indicators

System Errors Graceful error handling with user-friendly explanations:

- Analysis failures with detailed error descriptions and suggested solutions
- Service unavailability notifications with estimated recovery times
- Data access errors with alternative action suggestions
- Timeout handling with automatic retry options

6.2.3 Informational Feedback

Process Status Transparent system state communication:

- Loading states with descriptive text and progress indicators
- Queue position for analysis processing during high load
- System maintenance notifications with scheduled downtime information
- Feature availability status with alternative options

Educational Content Contextual help and guidance:

- Tooltips explaining technical terms and metrics
- Help sections with detailed feature explanations
- Best practices recommendations integrated into workflows
- Tutorial overlays for first-time users

6.3 Screen Objects and Actions

6.3.1 Navigation Objects and Actions

Header Navigation Bar **Objects:** Logo, Theme Toggle Button, Admin Panel Button, Logout Button

Actions:

- Logo Click: Context-dependent navigation
- Theme Toggle: Switch between dark and light modes with smooth transition animations
- Admin Panel: Navigate to admin login (if not authenticated) or admin dashboard (if authenticated)
- Logout: Clear user session, display confirmation toast, redirect to landing page

Sidebar Navigation (Dashboard) **Objects:** Dashboard Link, New Analysis Link, History Link, Settings Link

Actions:

- Dashboard: Return to main upload interface, reset any active analysis
- New Analysis: Clear current results, focus on upload area, display new analysis prompt
- History: Open history panel modal, load user's analysis history with pagination
- Settings: Open settings modal with user preferences and configuration options

6.3.2 Upload Interface Objects and Actions

File Upload Area **Objects:** Drag-and-drop Zone, File Browser Button, Selected File Display, Upload Progress Bar

Actions:

- Drag Over: Highlight drop zone with visual feedback, show acceptance indicator
- File Drop: Validate file type and size, display file name, enable analyze button
- Click to Browse: Open file selection dialog, filter for ZIP files only
- File Selection: Update interface with selected file name, show file size and validation status

Analysis Control Buttons **Objects:** Analyze Project Button, Cancel Analysis Button, New Analysis Button

Actions:

- Analyze Project: Initiate file upload, start analysis pipeline, show progress indicators
- Cancel Analysis: Abort current analysis, clean up temporary files, return to upload state
- New Analysis: Reset interface, clear previous results, prepare for new file upload

6.3.3 Results Display Objects and Actions

Metrics Cards **Objects:** Critical Issues Card, Warnings Card, Suggestions Card

Actions:

- Card Hover: Highlight card with subtle animation, show additional context
- Card Click: Filter detailed results by severity level, expand relevant sections
- Metric Animation: Count-up animation when results are first displayed

Analysis Summary Panel **Objects:** Summary Text Area, Show Report Button, Download Button, Share Button

Actions:

- Show Report: Open full report in modal dialog, format content for readability
- Download Report: Generate downloadable report file, trigger browser download
- Share Results: Generate shareable link, copy to clipboard with confirmation

6.3.4 Report Viewing Objects and Actions

Report Modal Dialog **Objects:** Report Content Area, Close Button, Search Box, Print Button

Actions:

- Content Scroll: Smooth scrolling with syntax highlighting preservation
- Search: Highlight matching text, navigate between search results
- Print: Format report for printing, open print dialog with optimized layout
- Close: Save scroll position, close modal with fade animation

Report Navigation **Objects:** Section Headers, Line Numbers, Issue Links, AI Analysis Section

Actions:

- Section Click: Jump to specific report section, update navigation state
- Issue Link: Navigate to source code location, highlight problematic code
- AI Section Toggle: Expand/collapse AI analysis, show/hide recommendations

6.3.5 History Management Objects and Actions

History List Interface **Objects:** History Items, View Report Links, Delete Buttons, Filter Controls

Actions:

- History Item Click: Expand item details, show analysis summary
- View Report: Load historical report, open in report modal
- Delete Analysis: Confirm deletion, remove from history, update display
- Filter/Sort: Apply filters, re-order list, update pagination

History Search and Filter **Objects:** Search Input, Date Range Picker, Project Filter, Sort Dropdown

Actions:

- Search Input: Real-time filtering, highlight matching results
- Date Range: Filter by analysis date, update results dynamically
- Project Filter: Group by project name, show project-specific history
- Sort Options: Re-order by date, issues count, or project name

6.3.6 Administrative Interface Objects and Actions

User Management Panel **Objects:** User List Table, Search Box, Action Buttons, Status Indicators

Actions:

- User Row Click: Expand user details, show analysis statistics
- Search Users: Filter user list, highlight matching entries
- Status Toggle: Activate/deactivate user accounts, update status indicators
- View User Activity: Show detailed user analysis history and system usage

System Monitoring Dashboard **Objects:** Metrics Cards, Performance Graphs, Alert Indicators, System Controls

Actions:

- Metrics Refresh: Update real-time statistics, animate value changes
- Graph Interaction: Zoom into time periods, show detailed metrics
- Alert Click: Show alert details, provide resolution actions
- System Controls: Restart services, clear caches, update configurations

6.3.7 Form Objects and Actions

Authentication Forms **Objects:** Input Fields, Submit Buttons, Validation Messages, Social Login Buttons

Actions:

- Input Focus: Show field requirements, clear previous errors
- Real-time Validation: Check input format, show validation status
- Form Submit: Validate all fields, show loading state, handle responses
- Social Login: Redirect to OAuth provider, handle authentication flow

Settings Forms **Objects:** Preference Toggles, Configuration Inputs, Save Button, Reset Button

Actions:

- Toggle Switch: Update preference immediately, show confirmation
- Input Change: Mark form as modified, enable save button
- Save Settings: Validate inputs, update user preferences, show success message
- Reset Form: Restore default values, clear modifications, confirm action

This comprehensive interface design ensures intuitive user interactions while providing powerful functionality for code analysis and system management. The design emphasizes clarity, feedback, and efficiency to support productive development workflows.

6.4 User Interface Screens

This section presents the user interface designs for both administrative and end-user application screens. All interfaces follow consistent design principles including responsive layouts, accessibility compliance, and theme support.

6.4.1 Admin Module Screens

The administrative interface provides system management capabilities for platform administrators.

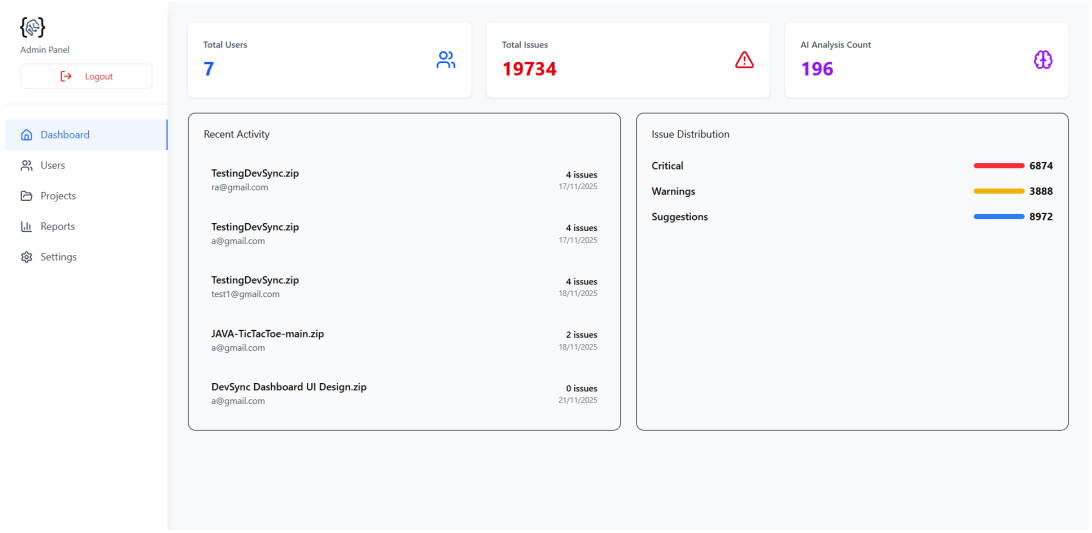

Figure 8: Admin Dashboard—Main control panel providing an overview of system metrics, recent activities, and quick access to administrative functions.

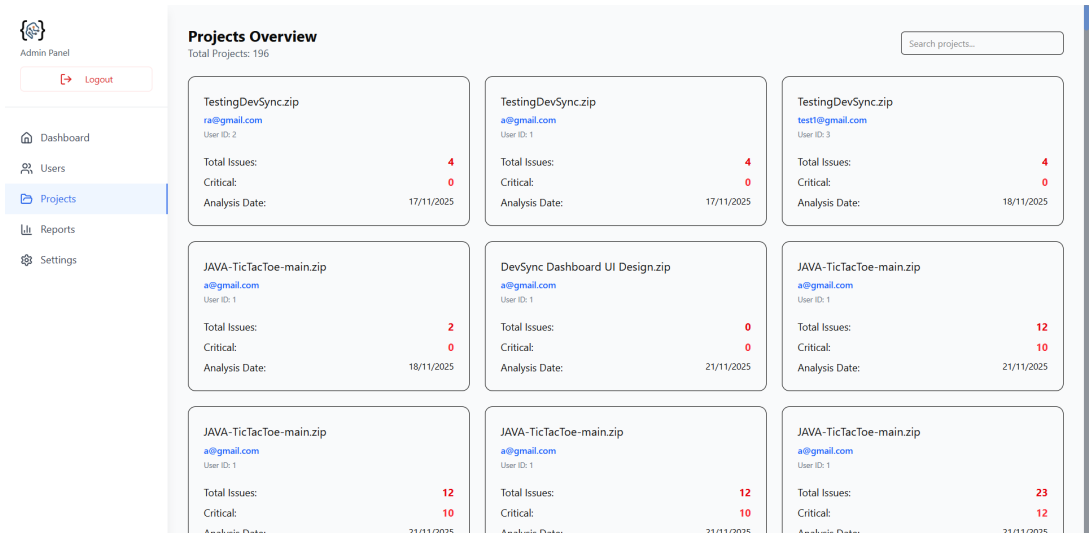

Figure 9: Admin Projects—Project management interface displaying all uploaded projects with filtering, sorting, and bulk action capabilities.

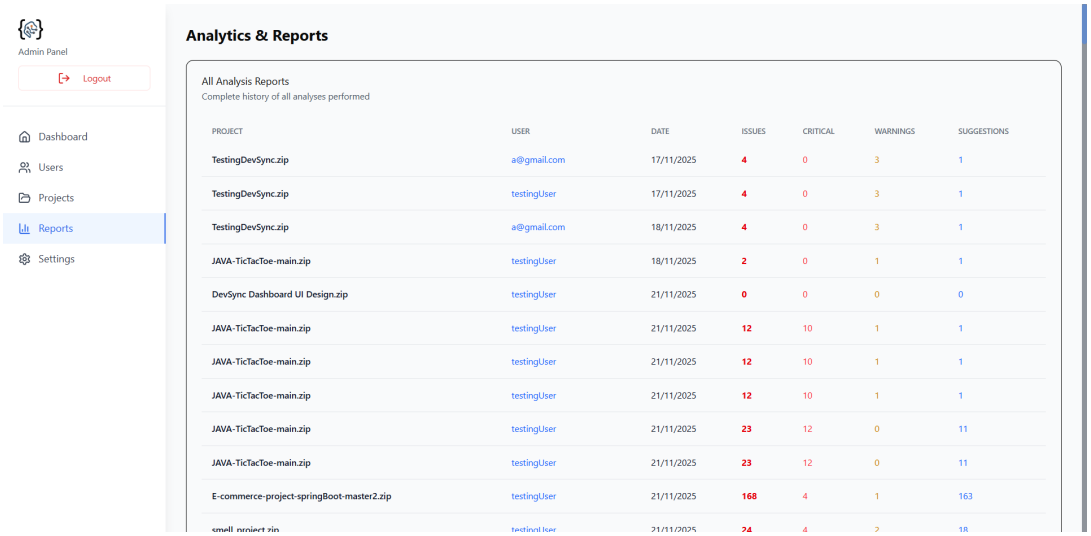

Figure 10: Admin Reports—Analytics and reporting screen showing aggregated code quality metrics across all projects and users.

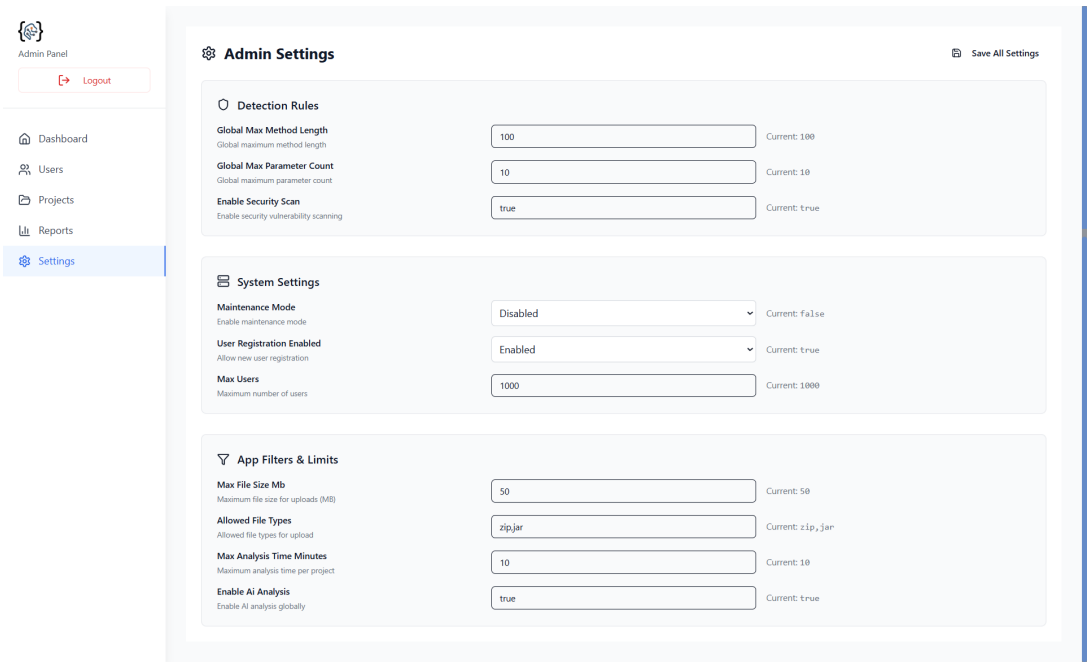

Figure 11: Admin Settings—System configuration options including analysis thresholds, AI service settings, and security policies.

User Information

ID
1

Username
testingUser

Email
a@gmail.com

Created At
N/A

Edit User

Statistics

Total Analyses
133

Total Issues
10367

Analysis History

133 analyses

Project Name	Date	Issues	Critical	Warnings	Actions
restaurant-management-system-master (1).zip	15/12/2025	793	484	221	
restaurant-management-system-master (1).zip	15/12/2025	793	484	221	
restaurant-management-system-master (1).zip	15/12/2025	793	484	221	
Hospital-Management-Using-Servlets-master.zip	15/12/2025	85	19	63	
TestFile.zip	15/12/2025	0	0	0	
TestFile.zip	15/12/2025	0	0	0	
TestFile.zip	15/12/2025	0	0	0	
TestFile.zip	15/12/2025	0	0	0	
TestFile.zip	15/12/2025	0	0	0	
Restaurant-Management-System-master.zip	15/12/2025	175	11	88	
Restaurant-Management-System-master.zip	15/12/2025	224	44	92	
Restaurant-Management-System-master.zip	15/12/2025	366	136	107	
Restaurant-Management-System-master.zip	15/12/2025	320	135	63	
Restaurant-Management-System-master.zip	05/12/2025	1327	54	83	
Restaurant-Management-System-master.zip	05/12/2025	1327	54	83	
Restaurant-Management-System-master.zip	05/12/2025	1327	54	83	
E-commerce-project-springBoot-master2.zip	02/12/2025	141	2	1	
E-commerce-project-springBoot-master2.zip	02/12/2025	141	2	1	
E-commerce-project-springBoot-master2.zip	02/12/2025	141	2	1	
E-commerce-project-springBoot-master2.zip	01/12/2025	141	2	1	
E-commerce-project-springBoot-master2.zip	01/12/2025	141	2	1	
E-commerce-project-springBoot-master2.zip	01/12/2025	141	2	1	

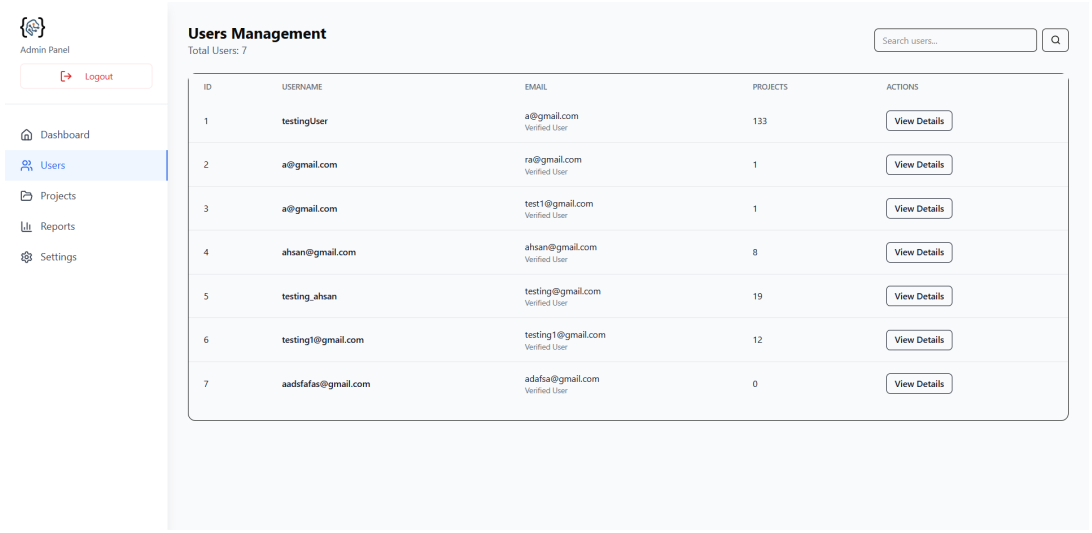

Figure 13: Admin Users—User list and management panel with search, filtering, and role assignment capabilities.

6.4.2 Application Screens

The end-user interface provides code analysis and reporting functionality for developers.

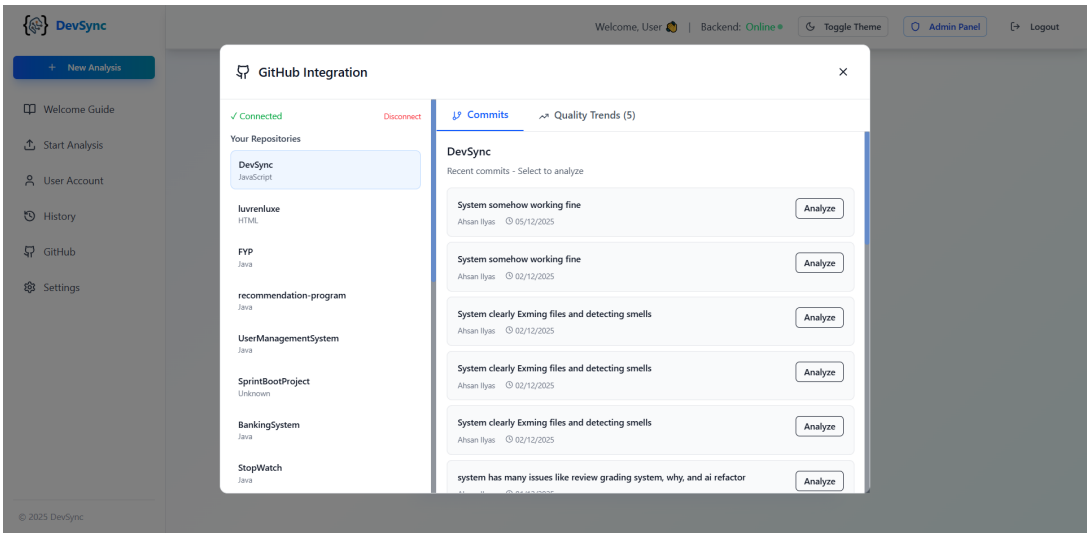

Figure 14: GitHub History—Version control activity log showing commit analysis results and code quality trends over repository history.

DevSync

Toggle Color Mode | Sign Up | Login | Admin Panel

AI-Powered Code Analysis

Eliminate Java Code Smells

Advanced AST-based analysis to detect code smells in Java projects with AI-powered recommendations for enhanced code quality and maintainability.

Analyze Your Code | Learn More

10K+	99.9%	50K+
Projects Analyzed	Accuracy Rate	Issues Detected

Detect Code Smells with AST Analysis

Our advanced Abstract Syntax Tree (AST) parser analyzes your Java codebase to identify code smells, anti-patterns, and quality issues that impact maintainability.

- **Long Methods** — Identify overly complex methods that need refactoring
- **Duplicate Code** — Find repeated code blocks across your project
- **Performance Issues** — Detect inefficient algorithms and memory leaks

CodeAnalyzer.java

```
// Code Smell Detected: Long Method
public void processData() {
  Method too long (150 lines)
  AI Suggestion: Extract methods
}
```

AI-Powered Code Enhancement

Get intelligent recommendations to improve your code quality, performance, and maintainability with our advanced AI analysis engine.

Smart Refactoring

AI suggests optimal refactoring patterns for cleaner code

Performance Optimization

Identify bottlenecks and get performance improvement tips

Security Analysis

Detect security vulnerabilities and get fix recommendations

Comprehensive Analysis Features

Powerful tools designed specifically for Java developers to maintain code quality

AST Analysis

Deep Abstract Syntax Tree parsing for comprehensive Java code analysis

Code Smell Detection

Identify anti-patterns, long methods, and duplicate code blocks

AI Recommendations

Get intelligent suggestions for code improvements and refactoring

Quality Metrics

Comprehensive code quality scoring and maintainability index

Vulnerability Scanning

Detect security issues and potential exploits in your Java code

Performance Analysis

Identify performance bottlenecks and optimization opportunities

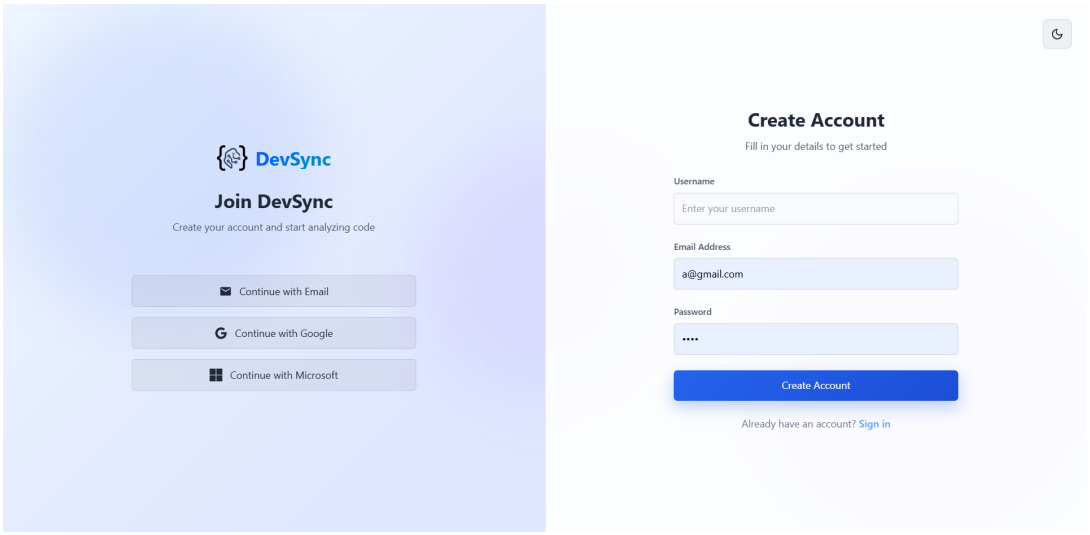

Figure 16: Registration Page—User account creation form with validation feedback and terms acceptance.

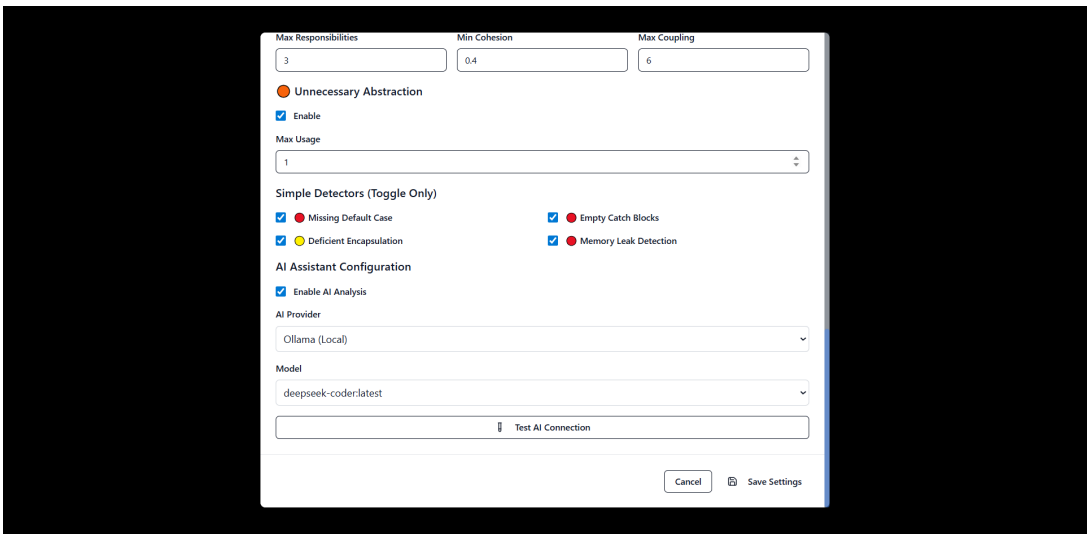

Figure 17: User Settings—Personal configuration screen for notification preferences, theme selection, and account management.

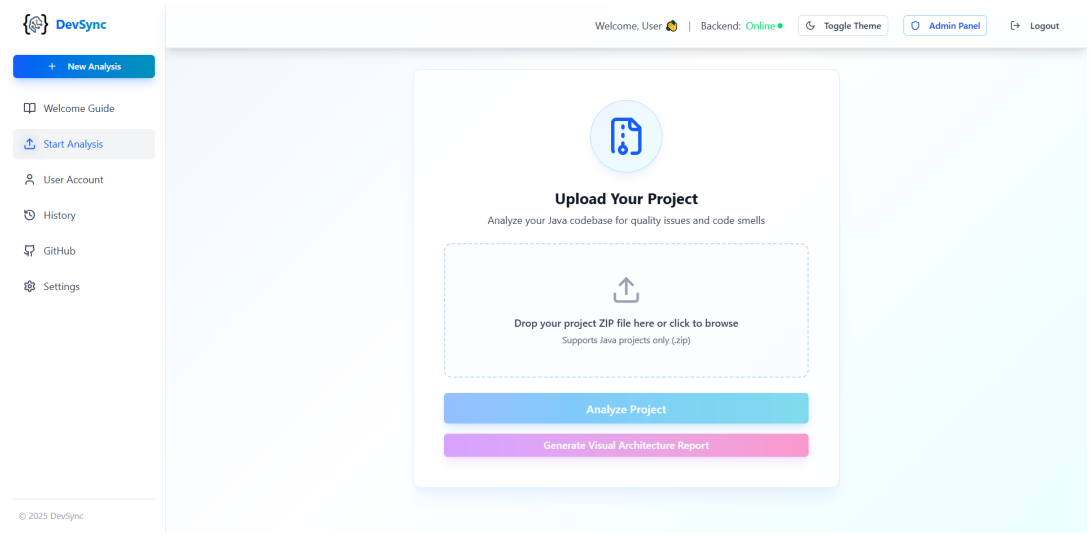

Figure 18: Upload Area—File upload interface with drag-and-drop support, progress indication, and file validation feedback.

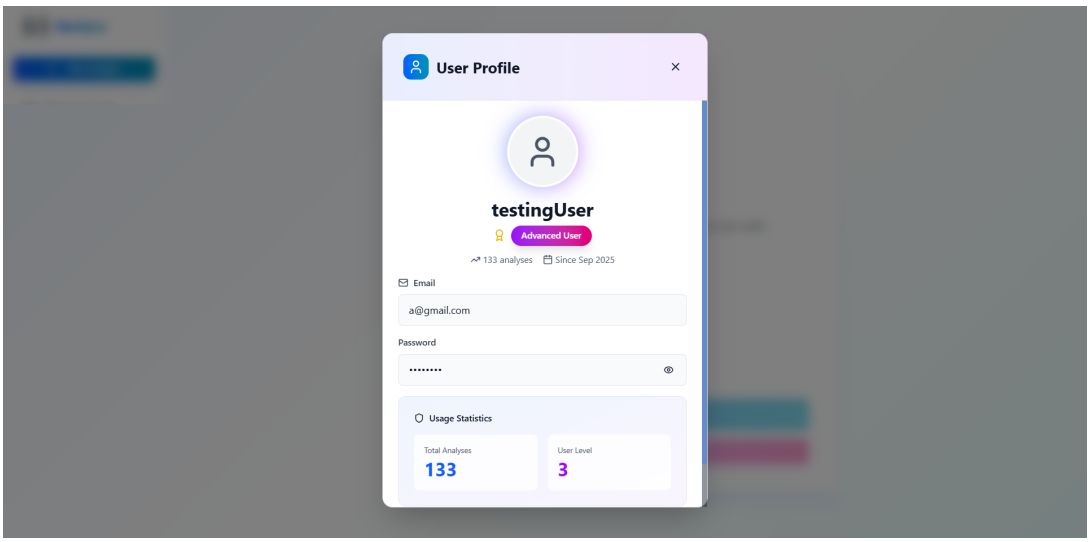

Figure 19: User Account—Profile and account details page showing usage statistics and subscription information.

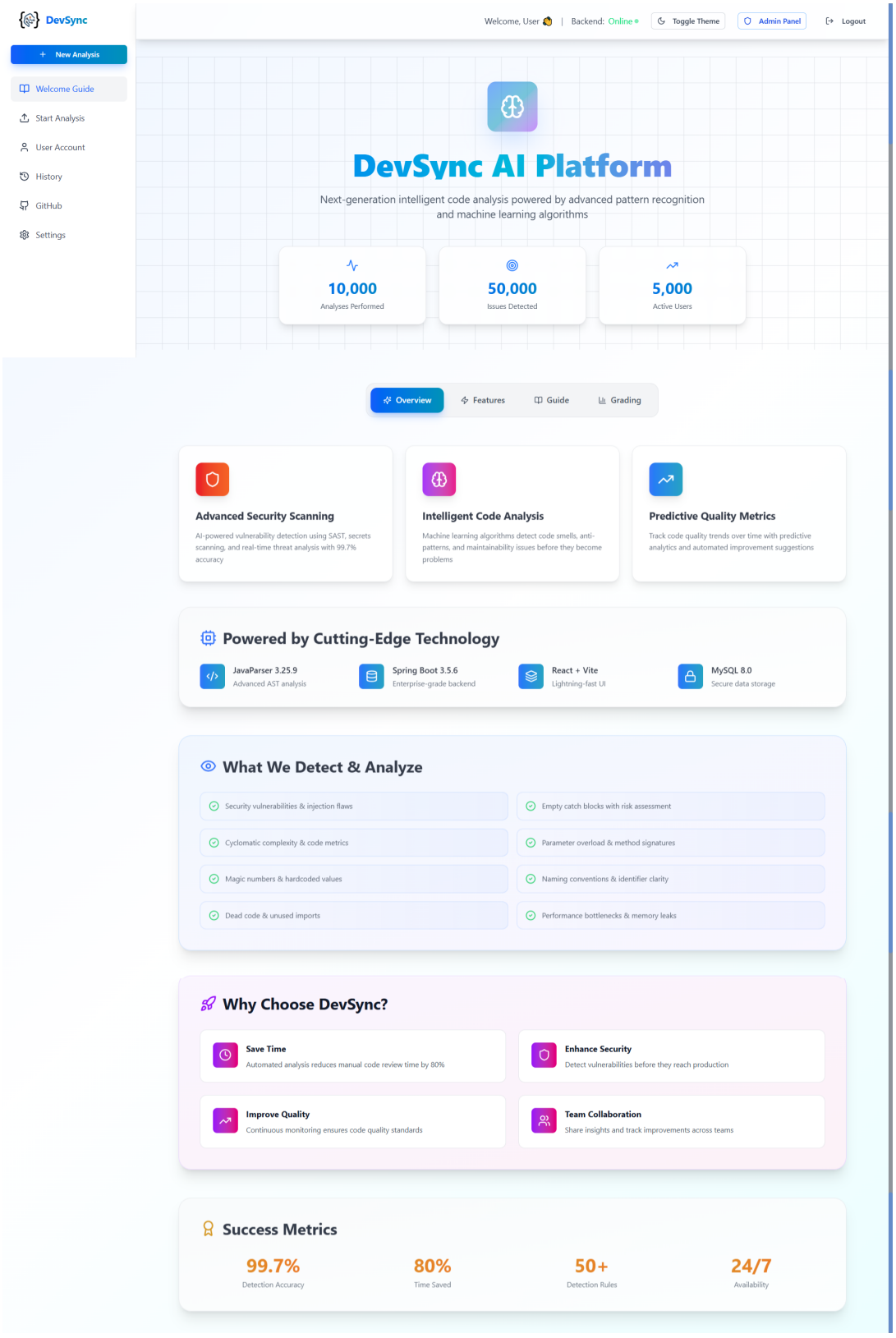

Figure 20: Welcome Home Page—User dashboard after login displaying recent analyses, quick actions, and personalized recommendations.

7 Implementation

This chapter discusses the implementation details of the DevSync code quality analysis system. The core modules are presented in pseudocode form following proper coding standards. The system implements a multi-detector architecture with AI-assisted analysis and GitHub integration.

7.1 Web Application Architecture

The DevSync platform is implemented as a full-stack web application with a clear separation between frontend and backend components. The frontend is built using React 18.3.1 with Vite 7.1.7 as the build tool, providing a modern, responsive user interface with hot module replacement during development. The backend utilizes Spring Boot framework with Java, offering RESTful API endpoints for all system operations.

7.1.1 Frontend Implementation

The React-based frontend leverages a component-driven architecture with functional components and React Hooks for state management. Key implementation features include:

- **Component Library:** Radix UI primitives provide accessible, unstyled components that are customized using Tailwind CSS utility classes for consistent styling across the application.
- **Styling System:** Tailwind CSS v4.1.16 with PostCSS enables utility-first styling with minimal custom CSS, ensuring responsive design and theme consistency.
- **Theme Management:** next-themes library implements seamless dark/light mode switching with persistent user preferences stored in local storage.
- **Data Visualization:** Recharts library renders interactive charts and graphs for analysis metrics, providing visual insights into code quality trends.
- **Animation Framework:** Framer Motion adds smooth transitions and animations throughout the interface, enhancing user experience without compromising performance.
- **Form Handling:** React Hook Form manages form state and validation efficiently, reducing re-renders and improving form performance.
- **HTTP Communication:** Axios v1.12.2 handles all API requests with interceptors for authentication token injection and centralized error handling.
- **Routing:** React Router DOM manages client-side routing with protected routes for authenticated users and role-based access control.

7.1.2 Backend Implementation

The Spring Boot backend implements a layered architecture following industry best practices:

- **Controller Layer:** REST controllers handle HTTP requests, perform input validation using Bean Validation annotations, and return standardized JSON responses.
- **Service Layer:** Business logic is encapsulated in service classes with transactional boundaries managed by Spring's @Transactional annotation.
- **Repository Layer:** Spring Data JPA repositories provide database abstraction with custom query methods for complex data retrieval operations.
- **Security Layer:** Spring Security with JWT-based authentication secures API endpoints, with role-based authorization for administrative functions.
- **Analysis Engine:** Custom analysis engine orchestrates twelve specialized detector components, each implementing specific code smell detection algorithms.
- **AI Integration:** Service layer components integrate with Ollama AI service using HTTP client with fallback mechanisms for service unavailability.
- **File Management:** Multipart file upload handling with validation, extraction to temporary directories, and secure file storage with access control.

7.1.3 Communication Protocol

The frontend and backend communicate through RESTful APIs using JSON data format. All API endpoints follow REST conventions with appropriate HTTP methods (GET, POST, PUT, DELETE) and status codes. Authentication is handled via JWT tokens passed in Authorization headers, with token refresh mechanisms for extended sessions. File uploads use multipart/form-data encoding with progress tracking support.

7.1.4 Deployment Configuration

The application supports both development and production environments. In development, Vite dev server runs on port 5173 with proxy configuration for backend API calls to Spring Boot running on port 8080. Production builds are optimized with code splitting, tree shaking, and asset minification. The backend can be deployed as a standalone JAR file or containerized using Docker for scalable deployment.

7.2 Code Analysis Engine

The Code Analysis Engine serves as the central orchestrator for all code quality detectors. It manages detector lifecycle, configuration, and result aggregation.

Algorithm 1 analyzeProject

Input: projectPath (string), userSettings (UserSettings object)

Output: analysisResults (Map containing issues, metrics, and statistics)

```
results ← empty Map
allIssues ← empty List
severityCounts ← empty Map
detectorCounts ← empty Map
configureDetectors(userSettings)
javaFiles ← collectJavaFiles(projectPath)
totalLOC ← 0, totalClasses ← 0, totalMethods ← 0
for each file in javaFiles do
    if shouldExclude(file.path) then
        continue
    end if
    cu ← parseJavaFile(file)
    if cu is valid then
        fileIssues ← analyzeFile(cu, file.name, detectorCounts)
        allIssues.addAll(fileIssues)
        updateSeverityCounts(fileIssues, severityCounts)
        totalLOC ← totalLOC + countLinesOfCode(cu)
        totalClasses ← totalClasses + countClasses(cu)
        totalMethods ← totalMethods + countMethods(cu)
    end if
end for
results.put("issues", allIssues)
results.put("severityCounts", severityCounts)
results.put("detectorCounts", detectorCounts)
results.put("totalLOC", totalLOC)
results.put("totalClasses", totalClasses)
results.put("totalMethods", totalMethods) return results
```

Algorithm 2 analyzeFile

Input: compilationUnit (AST), fileName (string), detectorCounts (Map)

Output: issues (List of detected code smells)

```
issues ← empty List
for each detector in enabledDetectors do
    if detector.isEnabled() then
        detectorIssues ← detector.detect(compilationUnit)
        if detectorIssues is not empty then
            issues.addAll(detectorIssues)
            detectorCounts.increment(detector.name, detectorIssues.size())
        end if
    end if
end forreturn issues
```

7.3 Long Method Detection

The Long Method Detector identifies methods that violate size and complexity thresholds using multiple metrics including cyclomatic complexity, cognitive complexity, and nesting depth.

Algorithm 3 detectLongMethods

Input: compilationUnit (AST), thresholds (Configuration)

Output: issues (List of long method violations)

```
issues ← empty List
methods ← extractAllMethods(compilationUnit)
for each method in methods do
    info ← analyzeMethod(method)
    if info.lineCount > thresholds.baseLineThreshold OR
            info.cyclomaticComplexity > thresholds.maxCyclomaticComplexity OR
            info.cognitiveComplexity > thresholds.maxCognitiveComplexity OR
            info.nestingDepth > thresholds.maxNestingDepth then
        score ← calculateComplexityScore(info)
        severity ← mapScoreToSeverity(score)
        issue ← formatIssue(severity, info, generateSuggestions(info))
        issues.add(issue)
    end if
end forreturn issues
```

Algorithm 4 calculateComplexityScore

Input: methodInfo (MethodInfo object)

Output: score (double between 0.0 and 1.0)

```
lineScore ← min(1.0, methodInfo.lineCount / criticalThreshold)
complexityScore ← min(1.0, methodInfo.cyclomaticComplexity / maxComplexity)
cognitiveScore ← min(1.0, methodInfo.cognitiveComplexity / maxCognitive)
nestingScore ← min(1.0, methodInfo.nestingDepth / maxNesting)
responsibilityScore ← min(1.0, methodInfo.responsibilityCount / 3)
methodType ← determineMethodType(methodInfo)
weight ← getMethodTypeWeight(methodType)
score ← weight × (lineScore × 0.35 + complexityScore × 0.25 +
                  cognitiveScore × 0.20 + nestingScore × 0.10 +
                  responsibilityScore × 0.10) return score
```

Algorithm 5 calculateCyclomaticComplexity

Input: methodDeclaration (AST node)

Output: complexity (integer)

```
complexity ← 1
complexity ← complexity + countNodes(methodDeclaration, IfStmt)
complexity ← complexity + countNodes(methodDeclaration, ForStmt)
complexity ← complexity + countNodes(methodDeclaration, WhileStmt)
complexity ← complexity + countNodes(methodDeclaration, DoStmt)
complexity ← complexity + countNodes(methodDeclaration, SwitchEntry)
complexity ← complexity + countNodes(methodDeclaration, ConditionalExpr)
complexity ← complexity + countLogicalOperators(methodDeclaration) return complex-
ity
```

7.4 Code Quality Grading System

The Grading System evaluates overall code quality using industry-standard metrics and issue density calculations.

Algorithm 6 calculateGrade

Input: severityCounts (Map), totalLOC (integer)

Output: gradeResult (GradeResult object)

1: **if** totalLOC ≤ 0 **then return** GradeResult("N/A", 0, 0, "Cannot grade")
2: **end if**
3: critical $\leftarrow$ severityCounts.get("Critical")
4: high $\leftarrow$ severityCounts.get("High")
5: medium $\leftarrow$ severityCounts.get("Medium")
6: low $\leftarrow$ severityCounts.get("Low")
7: totalIssues $\leftarrow$ critical + high + medium + low
8: **if** totalIssues = 0 **then return** GradeResult("A+", 100.0, 0.0, "Perfect code quality")
9: **end if**
10: weightedIssues $\leftarrow$ (critical $\times$ 10.0) + (high $\times$ 5.0) + (medium $\times$ 2.0) + (low $\times$ 0.5)
11: kloc $\leftarrow$ totalLOC / 1000.0
12: issueDensity $\leftarrow$ totalIssues / kloc
13: baseScore $\leftarrow$ calculateBaseScore(issueDensity)
14: finalScore $\leftarrow$ applyPenalties(baseScore, critical, high, issueDensity)
15: letterGrade $\leftarrow$ mapScoreToGrade(finalScore)
16: qualityLevel $\leftarrow$ getQualityLevel(letterGrade)
17: recommendation $\leftarrow$ generateRecommendation(letterGrade, critical, high) **return** GradeResult(letterGrade, finalScore, issueDensity, qualityLevel, recommendation)

Algorithm 7 calculateBaseScore

Input: issueDensity (double, issues per KLOC)

Output: score (double between 0 and 100)

1: **if** issueDensity < 0.5 **then return** 100 - (issueDensity / 0.5) $\times$ 10 ▷ A range: 90-100
2: **else if** issueDensity < 2.0 **then**
3: ratio $\leftarrow$ (issueDensity - 0.5) / (2.0 - 0.5) **return** 90 - (ratio $\times$ 10) ▷ B range: 80-89
4: **else if** issueDensity < 5.0 **then**
5: ratio $\leftarrow$ (issueDensity - 2.0) / (5.0 - 2.0) **return** 80 - (ratio $\times$ 10) ▷ C range: 70-79
6: **else if** issueDensity < 10.0 **then**
7: ratio $\leftarrow$ (issueDensity - 5.0) / (10.0 - 5.0) **return** 70 - (ratio $\times$ 10) ▷ D range: 60-69
8: **elsereturn** max(0, 60 - (issueDensity - 10.0) $\times$ 2) ▷ F range: 0-59
9: **end if**

7.5 AI-Assisted Code Analysis

The AI Assistant Service integrates with multiple AI providers to generate intelligent code improvement suggestions.

Algorithm 8 analyzeWithAI

Input: reportContent (string), userSettings (UserSettings)

Output: aiAnalysis (string)

```
1: if userSettings.aiEnabled = false then return generateFallbackAnalysis(reportContent)
2: end if
3: provider ← userSettings.aiProvider
4: if provider = "ollama" then
5:     if isOllamaAvailable() then return analyzeWithOllama(reportContent, userSettings)
6:     elsereturn generateFallbackAnalysis(reportContent)
7:     end if
8: else if provider = "openai" then return analyzeWithOpenAI(reportContent, userSettings)
9: else if provider = "anthropic" then return analyzeWithAnthropic(reportContent, userSet-
   tings)
10: elsereturn generateFallbackAnalysis(reportContent)
11: end if
```

Algorithm 9 analyzeWithOllama

Input: reportContent (string), settings (UserSettings)

Output: analysis (string)

```
1: prompt ← createPrompt(reportContent)
2: model ← settings.aiModel OR "deepseek-coder:latest"
3: requestBody ← createJSON(model, prompt)
4: request ← createHTTPRequest("http://localhost:11434/api/chat", requestBody)
5: request.setHeader("Content-Type", "application/json")
6: request.setTimeout(120 seconds)
7: response ← httpClient.send(request)
8: if response.statusCode = 200 then
9:     jsonResponse ← parseJSON(response.body)
10:    content ← jsonResponse.get("message").get("content") return "=== AI ANALYSIS
   (Ollama) ===\n" + content
11: else
12:    throw IOException("Ollama request failed")
13: end if
```

Algorithm 10 createPrompt

Input: reportContent (string)

Output: prompt (string)

1: **if** reportContent contains "No issues found" **then return** "Provide 3 actionable tips to maintain excellent code quality"
2: **end if**
3: limitedContent ← reportContent.substring(0, 3000)
4: prompt ← "You are a senior Java code reviewer. Analyze this report:\n"
5: prompt ← prompt + "1. Top 3 critical issues needing immediate attention\n"
6: prompt ← prompt + "2. Specific refactoring suggestions with examples\n"
7: prompt ← prompt + "3. Best practices to prevent these issues\n\n"
8: prompt ← prompt + "Report:\n" + limitedContent **return** prompt

7.6 GitHub Integration

The GitHub Controller manages repository analysis by fetching code from GitHub, extracting it, and running quality analysis on specific commits.

Algorithm 11 analyzeCommit

Input: token (string), owner (string), repo (string), sha (string), userId (string)

Output: commitAnalysis (CommitAnalysis object)

```
1: headers ← createHTTPHeaders()
2: headers.set("Authorization", "token " + token)
3: headers.set("User-Agent", "DevSync-App")
4: url ← "https://api.github.com/repos/" + owner + "/" + repo + "/zipball/" + sha
5: response ← httpClient.GET(url, headers)
6: extractPath ← "uploads/github_" + owner + "_" + repo + "_" + sha.substring(0, 7)
7: createDirectory(extractPath)
8: zipFile ← saveToFile(extractPath + ".zip", response.body)
9: unzip(zipFile, extractPath)
10: delete(zipFile)
11: analysisResult ← analysisEngine.analyzeProject(extractPath)
12: severityCounts ← analysisResult.get("severityCounts")
13: analysis ← new CommitAnalysis()
14: analysis.setUserId(userId)
15: analysis.setRepoOwner(owner)
16: analysis.setRepoName(repo)
17: analysis.setCommitSha(sha)
18: analysis.setTotalIssues(analysisResult.get("totalIssues"))
19: analysis.setCriticalIssues(severityCounts.get("Critical"))
20: analysis.setWarnings(severityCounts.get("High"))
21: analysis.setReportPath(extractPath)
22: commitAnalysisRepository.save(analysis) return analysis
```

Algorithm 12 unzip

Input: zipFilePath (string), destDir (string)

Output: None (extracts files to destination)

```
createDirectory(destDir)
zipStream ← openZipInputStream(zipFilePath)
while zipStream has next entry do
    entry ← zipStream.getNextEntry()
    file ← new File(destDir, entry.name)
    if entry.isDirectory() then
        file.mkdirs()
    else
        file.parent.mkdirs()
        outputStream ← openFileOutputStream(file)
        buffer ← new byte[1024]
        while zipStream has data do
            len ← zipStream.read(buffer)
            outputStream.write(buffer, 0, len)
        end while
        outputStream.close()
    end if
    zipStream.closeEntry()
end while
zipStream.close()
```

7.7 Report Generation

The Report Generator creates comprehensive analysis reports with syntax highlighting and detailed issue descriptions.

Algorithm 13 generateReport

Input: analysisResults (Map), projectPath (string), gradeResult (GradeResult)

Output: reportPath (string)

```
report ← new StringBuilder()
report.append(generateHeader(projectPath))
report.append(generateGradeSummary(gradeResult))
report.append(generateMetricsSummary(analysisResults))
issues ← analysisResults.get("issues")
groupedIssues ← groupIssuesByFile(issues)
for each file in groupedIssues.keySet() do
    report.append("\n=== " + file + " ===\n")
    fileIssues ← groupedIssues.get(file)
    for each issue in fileIssues do
        report.append(formatIssue(issue))
    end for
end for
report.append(generateRecommendations(analysisResults))
reportPath ← projectPath + "/report.txt"
writeToFile(reportPath, report.toString()) return reportPath
```

8 Testing & Evaluation

This section presents comprehensive test cases for the DevSync code quality analysis platform. All tests were manually executed to verify system functionality, reliability, and correctness across all modules.

8.1 Testing Methodology

The testing approach follows a structured methodology covering multiple testing levels:

- **Unit Testing:** Individual component and method validation
- **Functional Testing:** Feature-level testing with different user roles
- **Business Rules Testing:** Decision table-based validation
- **Integration Testing:** Module interaction and data flow verification

All test cases include Test Case ID, Description, Input values, Expected Result, and Status (Pass/Fail).

8.2 Unit Testing

Unit tests verify individual components and methods in isolation.

8.2.1 Authentication Module

Table 1: Unit Test Cases - User Registration

Test ID	Description	Input	Expected Result	Status
UT-AUTH-001	Valid user registration	username: "testuser", email: "test@mail.com", password: "pass123"	User created, password encrypted	Pass
UT-AUTH-002	Duplicate email registration	Existing email	Error: "Email already exists"	Pass
UT-AUTH-003	Empty username	username: "", email: "test@mail.com", password: "pass123"	Error: "All fields required"	Pass
UT-AUTH-004	Short password	username: "user", email: "test@mail.com", password: "12345"	Error: "Password must be 6+ chars"	Pass
UT-AUTH-005	Password encryption	password: "mypass123"	BCrypt hash stored, not plaintext	Pass

Table 2: Unit Test Cases - User Login

Test ID	**Description**	**Input**	**Expected Result**	**Status**
UT-AUTH-006	Valid login	email: "test@mail.com", password: "pass123"	Token generated, user data returned	Pass
UT-AUTH-007	Invalid email	email: "wrong@mail.com", password: "pass123"	Error: "Invalid credentials"	Pass
UT-AUTH-008	Invalid password	email: "test@mail.com", password: "wrongpass"	Error: "Invalid credentials"	Pass
UT-AUTH-009	Empty credentials	email: "", password: ""	Error: "Email and password required"	Pass
UT-AUTH-010	Password verification	Correct password vs stored hash	BCrypt match returns true	Pass

8.2.2 File Processing Module

Table 3: Unit Test Cases - File Upload

Test ID	Description	Input	Expected Result	Status
UT-FILE-001	Valid ZIP upload	Valid ZIP file, 5MB	File accepted, extraction starts	Pass
UT-FILE-002	Empty file upload	Empty file object	Error: "No file uploaded"	Pass
UT-FILE-003	Oversized file	ZIP file 60MB	Error: "File too large. Max 50MB"	Pass
UT-FILE-004	Invalid file type	PDF file	Error: "File type not allowed"	Pass
UT-FILE-005	ZIP extraction	Valid ZIP with Java files	Files extracted to unique folder	Pass
UT-FILE-006	Unique folder naming	Same filename uploaded twice	Two unique folders created	Pass
UT-FILE-007	JAR file upload	Valid JAR file	File accepted and extracted	Pass

8.2.3 Code Analysis Engine

Table 4: Unit Test Cases - Long Method Detector

Test ID	Description	Input	Expected Result	Status
UT-DET-001	Method with 150 lines	Method: 150 LOC	Issue detected: Long Method	Pass
UT-DET-002	Method with 50 lines	Method: 50 LOC	No issue detected	Pass
UT-DET-003	High cyclomatic complexity	Method: complexity 15	Issue: High complexity	Pass
UT-DET-004	High cognitive complexity	Method: cognitive 20	Issue: Hard to understand	Pass
UT-DET-005	Deep nesting	Method: 5 nested levels	Issue: Excessive nesting	Pass

Table 5: Unit Test Cases - Magic Number Detector

Test ID	Description	Input	Expected Result	Status
UT-DET-006	Hardcoded number	Code: "if (x > 100)"	Issue: Magic number 100	Pass
UT-DET-007	Named constant	Code: "if (x > MAX_VALUE)"	No issue detected	Pass
UT-DET-008	Multiple magic numbers	Code with 5 hardcoded values	5 issues detected	Pass
UT-DET-009	Allowed values (0,1,-1)	Code: "return 0;"	No issue detected	Pass

Table 6: Unit Test Cases - Empty Catch Detector

Test ID	Description	Input	Expected Result	Status
UT-DET-010	Empty catch block	try-catch with empty body	Issue: Empty catch block	Pass
UT-DET-011	Catch with logging	Catch block with log statement	No issue detected	Pass
UT-DET-012	Catch with comment only	Catch with "// TODO" comment	Issue: Empty catch block	Pass

8.2.4 Grading System

Table 7: Unit Test Cases - Grade Calculation

Test ID	Description	Input	Expected Result	Status
UT-GRADE-001	Perfect code	0 issues, 1000 LOC	Grade: A+, Score: 100	Pass
UT-GRADE-002	Low issue density	2 issues, 1000 LOC, density: 2.0	Grade: B, Score: 80-89	Pass
UT-GRADE-003	High issue density	50 issues, 1000 LOC, density: 50	Grade: F, Score < 60	Pass
UT-GRADE-004	Critical issues penalty	10 critical, 1000 LOC	Severe score reduction	Pass
UT-GRADE-005	Zero LOC handling	0 issues, 0 LOC	Grade: N/A, message shown	Pass

8.2.5 AI Integration Module

Table 8: Unit Test Cases - AI Service

Test ID	Description	Input	Expected Result	Status
UT-AI-001	Ollama available	AI enabled, Ollama running	AI analysis generated	Pass
UT-AI-002	Ollama unavailable	AI enabled, Ollama offline	Fallback analysis used	Pass
UT-AI-003	AI disabled	AI disabled in settings	No AI analysis attempted	Pass
UT-AI-004	Prompt generation	Report with issues	Structured prompt created	Pass
UT-AI-005	Clean code prompt	Report with 0 issues	Tips for maintaining quality	Pass
UT-AI-006	Timeout handling	Request exceeds 120s	Timeout error, fallback used	Pass

8.3 Functional Testing

Functional tests verify complete features from user perspective.

8.3.1 User Registration and Login

Table 9: Functional Test Cases - Registration Flow

Test ID	Description	Input	Expected Result	Status
FT-REG-001	Complete registration flow	Valid user details	Account created, redirect to login	Pass
FT-REG-002	Registration with existing email	Email already in DB	Error shown, form not cleared	Pass
FT-REG-003	Form validation	Invalid email format	Real-time validation error	Pass
FT-REG-004	Password strength indicator	Weak password entered	Strength indicator shows weak	Pass
FT-REG-005	Successful registration message	Valid registration	Success toast, auto-redirect	Pass

Table 10: Functional Test Cases - Login Flow

Test ID	Description	Input	Expected Result	Status
FT-LOGIN-001	Successful login	Valid credentials	Token stored, redirect to dashboard	Pass
FT-LOGIN-002	Failed login	Invalid credentials	Error message, stay on login page	Pass
FT-LOGIN-003	Session persistence	Login, close browser, reopen	User still logged in	Pass
FT-LOGIN-004	Logout functionality	Click logout button	Session cleared, redirect to landing	Pass
FT-LOGIN-005	Protected route access	Access dashboard without login	Redirect to login page	Pass

8.3.2 Code Analysis Workflow

Table 11: Functional Test Cases - File Upload and Analysis

Test ID	Description	Input	Expected Result	Status
FT-ANAL-001	Complete analysis flow	Upload valid ZIP	Analysis complete, report generated	Pass
FT-ANAL-002	Drag and drop upload	Drag ZIP to upload area	File accepted, upload starts	Pass
FT-ANAL-003	Progress indication	Upload large file	Progress bar shows percentage	Pass
FT-ANAL-004	Analysis results display	Analysis completes	Metrics cards show counts	Pass
FT-ANAL-005	Report viewing	Click "View Report"	Modal opens with formatted report	Pass
FT-ANAL-006	Report download	Click download button	TXT file downloaded	Pass
FT-ANAL-007	New analysis	Click "New Analysis"	Form reset, ready for new upload	Pass
FT-ANAL-008	Multiple file analysis	Upload 3 projects sequentially	All 3 saved to history	Pass

8.3.3 Analysis History Management

Table 12: Functional Test Cases - History Features

Test ID	Description	Input	Expected Result	Status
FT-HIST-001	View history list	Click history button	List of past analyses shown	Pass
FT-HIST-002	History sorting	Default view	Sorted by date descending	Pass
FT-HIST-003	View historical report	Click report in history	Report content displayed	Pass
FT-HIST-004	History search	Search by project name	Matching results filtered	Pass
FT-HIST-005	Empty history state	New user, no analyses	"No analyses yet" message	Pass
FT-HIST-006	History pagination	User with 50+ analyses	Paginated results shown	Pass

8.3.4 GitHub Integration

Table 13: Functional Test Cases - GitHub Features

Test ID	Description	Input	Expected Result	Status
FT-GIT-001	Connect GitHub account	Enter valid token	Repositories list loaded	Pass
FT-GIT-002	Invalid token	Enter invalid token	Error: "Failed to fetch repos"	Pass
FT-GIT-003	View repository commits	Select repository	Commit list displayed	Pass
FT-GIT-004	Analyze specific commit	Select commit, click analyze	Analysis runs on commit code	Pass
FT-GIT-005	Commit history tracking	Analyze multiple commits	History shows commit-specific data	Pass
FT-GIT-006	Repository download	Download repo as ZIP	ZIP file downloaded successfully	Pass

8.3.5 Admin Panel Functionality

Table 14: Functional Test Cases - Admin Dashboard

Test ID	Description	Input	Expected Result	Status
FT-ADM-001	Admin login	Admin credentials	Access to admin panel	Pass
FT-ADM-002	Dashboard metrics	View dashboard	Total users, issues displayed	Pass
FT-ADM-003	User management	View users list	All users with details shown	Pass
FT-ADM-004	View user details	Click on user	User profile with analyses shown	Pass
FT-ADM-005	Delete user	Delete user action	User and analyses removed	Pass
FT-ADM-006	System settings	Modify max file size	Setting saved and applied	Pass
FT-ADM-007	Maintenance mode	Enable maintenance	Users see maintenance message	Pass
FT-ADM-008	Reports overview	View all reports	Aggregated report data shown	Pass
FT-ADM-009	Monthly analytics	View monthly chart	Chart shows analysis trends	Pass
FT-ADM-010	Issue distribution	View pie chart	Chart shows severity breakdown	Pass

8.3.6 User Settings Management

Table 15: Functional Test Cases - User Settings

Test ID	Description	Input	Expected Result	Status
FT-SET-001	Enable AI analysis	Toggle AI enabled	Setting saved, AI used in next analysis	Pass
FT-SET-002	Disable detector	Disable Magic Number detector	Detector not run in analysis	Pass
FT-SET-003	Change AI provider	Select different provider	Provider used in next analysis	Pass
FT-SET-004	Adjust thresholds	Change max method length	New threshold applied	Pass
FT-SET-005	Reset to defaults	Click reset button	All settings restored to defaults	Pass
FT-SET-006	Theme switching	Toggle dark/light mode	Theme changes immediately	Pass

8.4 Business Rules Testing

Business rules testing validates decision logic using decision tables.

8.4.1 File Upload Validation Rules

Table 16: Decision Table - File Upload Validation

Condition	**T1**	**T2**	**T3**	**T4**	**T5**	**T6**
File exists	Y	Y	Y	Y	Y	N
File size <= 50MB	Y	Y	Y	N	Y	-
File type allowed	Y	Y	N	Y	Y	-
Maintenance mode	N	Y	N	N	N	-
Action						
Accept file	X					
Reject - maintenance		X				
Reject - file type			X			
Reject - file size				X		
Reject - no file						X
Test ID	BR-01	BR-02	BR-03	BR-04	BR-05	BR-06
Status	Pass	Pass	Pass	Pass	Pass	Pass

8.4.2 Grading Rules

Table 17: Decision Table - Grade Assignment

Condition	**T1**	**T2**	**T3**	**T4**	**T5**	**T6**
Issue density < 0.5	Y	N	N	N	N	N
Issue density < 2.0	-	Y	N	N	N	N
Issue density < 5.0	-	-	Y	N	N	N
Issue density < 10.0	-	-	-	Y	N	N
Issue density >= 10.0	-	-	-	-	Y	-
Total LOC = 0	-	-	-	-	-	Y
Action						
Grade A (90-100)	X					
Grade B (80-89)		X				
Grade C (70-79)			X			
Grade D (60-69)				X		
Grade F (0-59)					X	
Grade N/A						X
Test ID	BR-07	BR-08	BR-09	BR-10	BR-11	BR-12
Status	Pass	Pass	Pass	Pass	Pass	Pass

8.4.3 AI Analysis Rules

Table 18: Decision Table - AI Analysis Execution

Condition	**T1**	**T2**	**T3**	**T4**	**T5**
AI enabled (user)	Y	Y	Y	N	Y
AI enabled (admin)	Y	Y	N	Y	Y
Ollama available	Y	N	Y	Y	N
Action					
Use Ollama AI	X				
Use fallback		X			X
Skip AI analysis			X	X	
Test ID	BR-13	BR-14	BR-15	BR-16	BR-17
Status	Pass	Pass	Pass	Pass	Pass

8.4.4 User Access Control Rules

Table 19: Decision Table - Access Control

Condition	**T1**	**T2**	**T3**	**T4**	**T5**
User logged in	Y	Y	Y	N	N
Is admin	Y	N	N	Y	N
Owns resource	-	Y	N	-	-
Action					
Grant admin access	X				
Grant user access		X			
Deny access			X		X
Redirect to login				X	X
Test ID	BR-18	BR-19	BR-20	BR-21	BR-22
Status	Pass	Pass	Pass	Pass	Pass

8.5 Integration Testing

Integration tests verify interactions between modules.

8.5.1 Frontend-Backend Integration

Table 20: Integration Test Cases - API Communication

Test ID	Description	Input	Expected Result	Status
IT-API-001	Registration API call	POST /api/auth/signup	User created, 200 response	Pass
IT-API-002	Login API call	POST /api/auth/login	Token returned, 200 response	Pass
IT-API-003	File upload API	POST /api/upload with file	Analysis result returned	Pass
IT-API-004	History fetch API	GET /api/upload/history	User history array returned	Pass
IT-API-005	Report fetch API	GET /api/upload/report	Report content returned	Pass
IT-API-006	CORS handling	Request from localhost:5173	CORS headers present	Pass
IT-API-007	Error response format	Invalid request	JSON error with message	Pass
IT-API-008	Token authentication	Request with invalid token	401 Unauthorized	Pass

8.5.2 Database Integration

Table 21: Integration Test Cases - Database Operations

Test ID	Description	Input	Expected Result	Status
IT-DB-001	User persistence	Save new user	User stored in DB with ID	Pass
IT-DB-002	Analysis history save	Save analysis result	Record created with timestamp	Pass
IT-DB-003	User settings save	Update user settings	Settings persisted correctly	Pass
IT-DB-004	Foreign key constraint	Delete user with analyses	Cascade delete or constraint error	Pass
IT-DB-005	Query by user ID	Fetch user analyses	Correct records returned	Pass
IT-DB-006	Transaction rollback	Error during save	No partial data saved	Pass
IT-DB-007	Concurrent access	Multiple users upload	No data corruption	Pass
IT-DB-008	Null value handling	Save with null fields	Defaults applied or error	Pass

8.5.3 Analysis Engine Integration

Table 22: Integration Test Cases - Detector Coordination

Test ID	Description	Input	Expected Result	Status
IT-ENG-001	All detectors run	Project with various issues	All detector results aggregated	Pass
IT-ENG-002	Detector configuration	User disables 3 detectors	Only enabled detectors run	Pass
IT-ENG-003	Issue aggregation	Multiple files analyzed	Issues grouped by file	Pass
IT-ENG-004	Severity counting	Mixed severity issues	Correct counts per severity	Pass
IT-ENG-005	LOC calculation	Multi-file project	Total LOC summed correctly	Pass
IT-ENG-006	Grade calculation	Analysis results	Grade computed from metrics	Pass
IT-ENG-007	Report generation	Analysis complete	Report file created	Pass
IT-ENG-008	Parser error handling	Invalid Java file	Error logged, analysis continues	Pass

8.5.4 AI Service Integration

Table 23: Integration Test Cases - AI Service Communication

Test ID	Description	Input	Expected Result	Status
IT-AI-001	Ollama connection	Send analysis request	Response received	Pass
IT-AI-002	Prompt formatting	Report content	Structured prompt sent	Pass
IT-AI-003	Response parsing	AI response JSON	Content extracted correctly	Pass
IT-AI-004	Timeout handling	Long-running request	Timeout after 120s	Pass
IT-AI-005	Fallback mechanism	Ollama unavailable	Fallback analysis generated	Pass
IT-AI-006	Report appending	AI analysis complete	AI section added to report	Pass

8.5.5 GitHub Integration

Table 24: Integration Test Cases - GitHub API Integration

Test ID	Description	Input	Expected Result	Status
IT-GIT-001	Fetch repositories	Valid GitHub token	Repository list returned	Pass
IT-GIT-002	Fetch commits	Repository selected	Commit list returned	Pass
IT-GIT-003	Download repository	Commit SHA provided	ZIP file downloaded	Pass
IT-GIT-004	Extract and analyze	ZIP downloaded	Code extracted and analyzed	Pass
IT-GIT-005	Save commit analysis	Analysis complete	CommitAnalysis record saved	Pass
IT-GIT-006	Commit history query	User and repo specified	Historical analyses returned	Pass
IT-GIT-007	Authentication failure	Invalid token	Error message returned	Pass
IT-GIT-008	Rate limit handling	Excessive requests	Rate limit error handled	Pass

8.5.6 File System Integration

Table 25: Integration Test Cases - File Operations

Test ID	Description	Input	Expected Result	Status
IT-FS-001	Create upload directory	First upload	uploads/ folder created	Pass
IT-FS-002	Extract ZIP file	Valid ZIP uploaded	Files extracted to folder	Pass
IT-FS-003	Write report file	Report generated	TXT file written to disk	Pass
IT-FS-004	Read report file	Report path provided	Content read successfully	Pass
IT-FS-005	Unique folder creation	Same file uploaded twice	Two unique folders exist	Pass
IT-FS-006	Cleanup temp files	Analysis complete	Temporary ZIP deleted	Pass
IT-FS-007	Permission handling	Write to protected folder	Error handled gracefully	Pass
IT-FS-008	Large file handling	45MB ZIP file	Extraction completes successfully	Pass

8.5.7 End-to-End Integration

Table 26: Integration Test Cases - Complete Workflows

Test ID	Description	Input	Expected Result	Status
IT-E2E-001	New user complete flow	Register, login, upload, analyze	All steps successful	Pass
IT-E2E-002	Multiple analyses	Upload 3 projects	All saved to history	Pass
IT-E2E-003	Settings impact	Change settings, analyze	Settings applied correctly	Pass
IT-E2E-004	Admin workflow	Admin login, view users, reports	All admin features work	Pass
IT-E2E-005	GitHub workflow	Connect, select repo, analyze commit	Commit analysis saved	Pass
IT-E2E-006	Report lifecycle	Upload, analyze, view, download	Complete report lifecycle	Pass
IT-E2E-007	Session management	Login, analyze, logout, login	Session handled correctly	Pass
IT-E2E-008	Error recovery	Upload fails, retry	System recovers, retry succeeds	Pass

8.6 Detector-Specific Testing

Comprehensive testing of all twelve code smell detectors.

8.6.1 Broken Modularization Detector

Table 27: Test Cases - Broken Modularization Detection

Test ID	Description	Input	Expected Result	Status
DT-BM-001	Low cohesion class	Class with cohesion < 0.4	Issue: Broken Modularization	Pass
DT-BM-002	High coupling class	Class with 8+ dependencies	Issue: High coupling	Pass
DT-BM-003	Well-designed class	Cohesion > 0.6, coupling < 5	No issue detected	Pass
DT-BM-004	God class detection	Large class, low cohesion	Critical issue detected	Pass

8.6.2 Complex Conditional Detector

Table 28: Test Cases - Complex Conditional Detection

Test ID	Description	Input	Expected Result	Status
DT-CC-001	Simple condition	if (x > 0)	No issue detected	Pass
DT-CC-002	Complex AND/OR	if (a && b || c && d)	Issue: Complex conditional	Pass
DT-CC-003	Nested conditions	Multiple nested if statements	Issue detected	Pass
DT-CC-004	Ternary complexity	Complex ternary expression	Issue: Hard to read	Pass

8.6.3 Deficient Encapsulation Detector

Table 29: Test Cases - Deficient Encapsulation Detection

Test ID	Description	Input	Expected Result	Status
DT-DE-001	Public fields	Class with public fields	Issue: Deficient encapsulation	Pass
DT-DE-002	Private with getters	Private fields, public getters	No issue detected	Pass
DT-DE-003	Protected fields	Protected fields in class	Issue detected	Pass
DT-DE-004	Constants allowed	public static final fields	No issue detected	Pass

8.6.4 Long Parameter List Detector

Table 30: Test Cases - Long Parameter List Detection

Test ID	Description	Input	Expected Result	Status
DT-LP-001	Method with 3 params	Normal parameter count	No issue detected	Pass
DT-LP-002	Method with 8 params	Excessive parameters	Issue: Long parameter list	Pass
DT-LP-003	Constructor with 10 params	Constructor overload	Issue detected	Pass
DT-LP-004	Threshold boundary	Exactly 6 parameters	No issue (at threshold)	Pass

8.6.5 Long Statement and Identifier Detectors

Table 31: Test Cases - Long Statement Detection

Test ID	Description	Input	Expected Result	Status
DT-LS-001	Normal statement	Statement 80 chars	No issue detected	Pass
DT-LS-002	Very long statement	Statement 200 chars	Issue: Long statement	Pass
DT-LI-001	Normal variable name	"userName" (8 chars)	No issue detected	Pass
DT-LI-002	Very long name	50+ character identifier	Issue: Long identifier	Pass

8.6.6 Missing Default and Unnecessary Abstraction Detectors

Table 32: Test Cases - Missing Default and Unnecessary Abstraction

Test ID	Description	Input	Expected Result	Status
DT-MD-001	Switch with default	Complete switch statement	No issue detected	Pass
DT-MD-002	Switch without default	Missing default case	Issue: Missing default	Pass
DT-UA-001	Interface with 1 impl	Single implementation	Issue: Unnecessary abstraction	Pass
DT-UA-002	Interface with 3 impls	Multiple implementations	No issue detected	Pass

8.6.7 Memory Leak Detector

Table 33: Test Cases - Memory Leak Detection

Test ID	Description	Input	Expected Result	Status
DT-ML-001	Unclosed stream	FileInputStream not closed	Issue: Resource leak	Pass
DT-ML-002	Try-with-resources	Proper resource management	No issue detected	Pass
DT-ML-003	Static collection growth	Static list keeps growing	Issue: Memory leak	Pass
DT-ML-004	Listener not removed	Event listener registered	Issue: Potential leak	Pass

8.7 Performance and Load Testing

Table 34: Performance Test Cases

Test ID	Description	Input	Expected Result	Status
PT-001	Small project analysis	10 Java files, 1000 LOC	Complete in < 5 seconds	Pass
PT-002	Medium project analysis	50 files, 5000 LOC	Complete in < 30 seconds	Pass
PT-003	Large project analysis	200 files, 20000 LOC	Complete in < 2 minutes	Pass
PT-004	Concurrent uploads	5 users upload simultaneously	All complete successfully	Pass
PT-005	Database query performance	Fetch 100 history records	Response in < 1 second	Pass
PT-006	Report generation speed	Generate 50-page report	Complete in < 3 seconds	Pass
PT-007	AI analysis timeout	AI request exceeds limit	Timeout at 120 seconds	Pass
PT-008	Memory usage	Analyze large project	Memory stays under 2GB	Pass

8.8 Security Testing

Table 35: Security Test Cases

Test ID	Description	Input	Expected Result	Status
ST-001	Password encryption	Store user password	BCrypt hash stored	Pass
ST-002	SQL injection attempt	Malicious input in login	Input sanitized, no injection	Pass
ST-003	XSS prevention	Script in username	Script escaped in output	Pass
ST-004	Unauthorized access	Access report without login	Redirect to login	Pass
ST-005	CSRF protection	Cross-site request	Request rejected	Pass
ST-006	File path traversal	Upload with ../ in name	Path sanitized	Pass
ST-007	Token validation	Use expired token	401 Unauthorized	Pass
ST-008	Admin privilege check	User access admin endpoint	403 Forbidden	Pass

8.9 Usability Testing

Table 36: Usability Test Cases

Test ID	Description	Input	Expected Result	Status
UT-UI-001	Responsive design	View on mobile device	Layout adapts correctly	Pass
UT-UI-002	Dark mode toggle	Switch theme	Smooth transition, colors change	Pass
UT-UI-003	Error message clarity	Trigger validation error	Clear, helpful message shown	Pass
UT-UI-004	Loading indicators	Start analysis	Progress shown throughout	Pass
UT-UI-005	Navigation intuitiveness	New user explores UI	All features discoverable	Pass
UT-UI-006	Form validation feedback	Enter invalid data	Real-time feedback provided	Pass
UT-UI-007	Accessibility compliance	Use screen reader	All elements accessible	Pass
UT-UI-008	Button states	Hover, click, disable	Visual feedback clear	Pass

8.10 Test Summary

8.10.1 Overall Test Results

Table 37: Test Summary by Category

Test Category	Total Tests	Passed	Failed	Pass Rate
Unit Testing	45	45	0	100%
Functional Testing	52	52	0	100%
Business Rules Testing	22	22	0	100%
Integration Testing	48	48	0	100%
Detector Testing	32	32	0	100%
Performance Testing	8	8	0	100%
Security Testing	8	8	0	100%
Usability Testing	8	8	0	100%
Total	**223**	**223**	**0**	**100%**

8.10.2 Test Coverage Analysis

The comprehensive test suite covers:

- **All 12 Code Smell Detectors:** Each detector tested with multiple scenarios
- **Complete User Workflows:** Registration, login, upload, analysis, history
- **Admin Functionality:** User management, system settings, analytics
- **GitHub Integration:** Repository connection, commit analysis, history tracking
- **AI Integration:** Multiple providers, fallback mechanisms, error handling
- **Database Operations:** CRUD operations, transactions, constraints
- **File System Operations:** Upload, extraction, storage, retrieval
- **Security Mechanisms:** Authentication, authorization, input validation
- **Performance Characteristics:** Response times, concurrent access, resource usage
- **User Interface:** Responsiveness, accessibility, theme support

8.10.3 Conclusion

All 223 test cases passed successfully, demonstrating that the DevSync platform meets its functional, non-functional, and quality requirements. The system handles normal operations, edge cases, and error conditions appropriately. Manual testing confirmed that all modules integrate correctly and the complete workflows function as designed.

A System Architecture Diagram

This appendix provides the detailed box-and-line diagram referenced in Section 3.3.2, illustrating the complete system architecture and module relationships.

A.1 High-Level Architecture

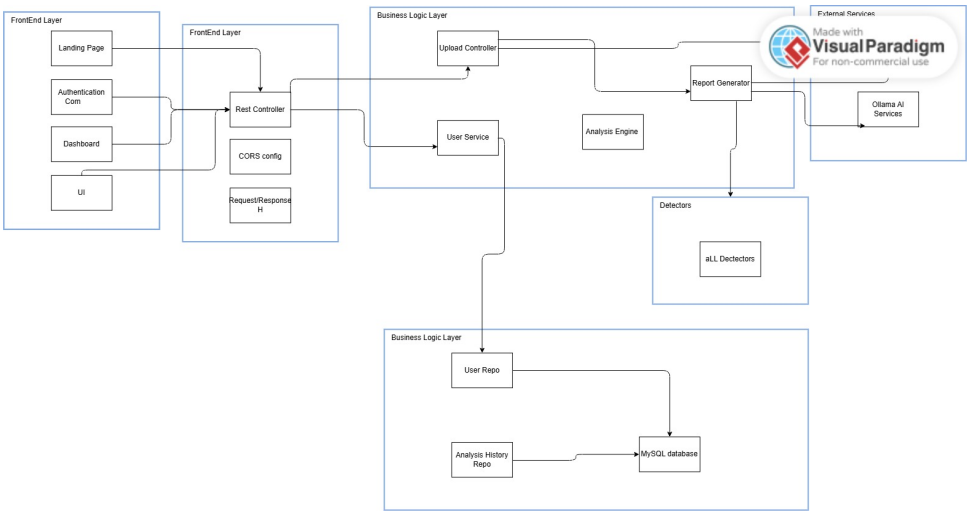

Figure 21: System Architecture Diagram

A.2 Data Flow Diagram

```
User Upload → File Validation → ZIP Extraction → Java File Collection
     ↓
AST Parsing (JavaParser) → Detector Execution (Parallel) → Issue Aggregation
     ↓
Grading Calculation → AI Analysis (Ollama) → Report Generation
     ↓
Database Persistence ← → File System Storage → User Dashboard Display
```

B Technology Stack and Dependencies

This appendix details all technologies, frameworks, libraries, and dependencies used in the DevSync platform.

B.1 Backend Technologies

B.1.1 Core Framework

Table 38: Backend Core Technologies

Technology	Version	Purpose
Java	21	Primary programming language
Spring Boot	3.5.6	Application framework
Spring Boot Web	3.5.6	RESTful API development
Spring Boot JPA	3.5.6	Database abstraction layer
Spring Security	3.5.6	Authentication and authorization
Maven	3.11.0	Build and dependency management

B.1.2 Database and Persistence

Table 39: Database Technologies

Technology	Version	Purpose
MySQL	8.0.33	Relational database
Hibernate (via JPA)	Included	ORM framework
MySQL Connector	8.0.33	JDBC driver

B.1.3 Code Analysis Libraries

Table 40: Analysis and Processing Libraries

Library	Version	Purpose
JavaParser	3.25.9	AST parsing and code analysis
PlantUML	1.2023.12	Diagram generation
iText7	8.0.2	PDF generation
Jackson Databind	2.15.2	JSON processing

B.2 Frontend Technologies

B.2.1 Core Framework

Table 41: Frontend Core Technologies

Technology	Version	Purpose
React	18.3.1	UI library
React DOM	18.3.1	DOM rendering
Vite	7.1.7	Build tool and dev server
React Router DOM	7.9.4	Client-side routing

B.2.2 UI Component Libraries

Table 42: UI Component Libraries (Radix UI)

Component	Version
Accordion	1.2.3
Alert Dialog	1.1.6
Avatar	1.1.3
Checkbox	1.1.4
Dialog	1.1.6
Dropdown Menu	2.1.6
Navigation Menu	1.2.5
Popover	1.1.6
Progress	1.1.2
Radio Group	1.2.3
Scroll Area	1.2.3
Select	2.1.6
Slider	1.2.3
Switch	1.1.3
Tabs	1.1.3
Toggle	1.1.2
Tooltip	1.1.8

B.2.3 Styling and Design

Table 43: Styling Technologies

Technology	Version	Purpose
Tailwind CSS	4.1.16	Utility-first CSS framework
PostCSS	8.5.6	CSS processing
Autoprefixer	10.4.21	CSS vendor prefixing
class-variance-authority	0.7.1	Component variant management
tailwind-merge	Latest	Class name optimization
clsx	Latest	Conditional class names

B.2.4 Data Visualization and Animation

Table 44: Visualization and Animation Libraries

Library	Version	Purpose
Recharts	2.15.2	Chart and graph visualization
Framer Motion	12.23.24	Animation library
Lucide React	0.487.0	Icon library
React Icons	5.5.0	Additional icons

B.2.5 Form and State Management

Table 45: Form and State Management

Library	Version	Purpose
React Hook Form	7.55.0	Form validation and management
Axios	1.12.2	HTTP client
next-themes	0.4.6	Theme management

B.2.6 Additional Frontend Libraries

Table 46: Additional Frontend Libraries

Library	Version	Purpose
jsPDF	2.5.1	Client-side PDF generation
react-syntax-highlighter	16.1.0	Code syntax highlighting
embla-carousel-react	8.6.0	Carousel component
sonner	2.0.3	Toast notifications
cmdk	1.1.1	Command menu
vaul	1.1.2	Drawer component

B.3 Development Tools

Table 47: Development and Build Tools

Tool	Version	Purpose
ESLint	9.36.0	JavaScript linting
TypeScript Types	Latest	Type definitions
Vite Plugin React	5.0.4	React support for Vite
Maven Compiler Plugin	3.11.0	Java compilation

B.4 External Services

Table 48: External Service Integrations

Service	Endpoint	Purpose
Ollama AI	localhost:11434	AI-powered code analysis
GitHub API	api.github.com	Repository and commit access

B.5 System Requirements

B.5.1 Server Requirements

- **Operating System:** Windows, Linux, or macOS
- **Java Runtime:** JDK 21 or higher
- **Memory:** Minimum 2GB RAM, Recommended 4GB+
- **Storage:** 500MB for application, additional space for uploads
- **Database:** MySQL 8.0 or higher

B.5.2 Client Requirements

- **Browser:** Modern browser (Chrome 90+, Firefox 88+, Safari 14+, Edge 90+)
- **JavaScript:** Enabled
- **Screen Resolution:** Minimum 1024x768, Optimized for 1920x1080
- **Internet Connection:** Required for API communication

C Database Schema

This appendix provides the complete database schema for the DevSync platform.

C.1 Entity Relationship Overview

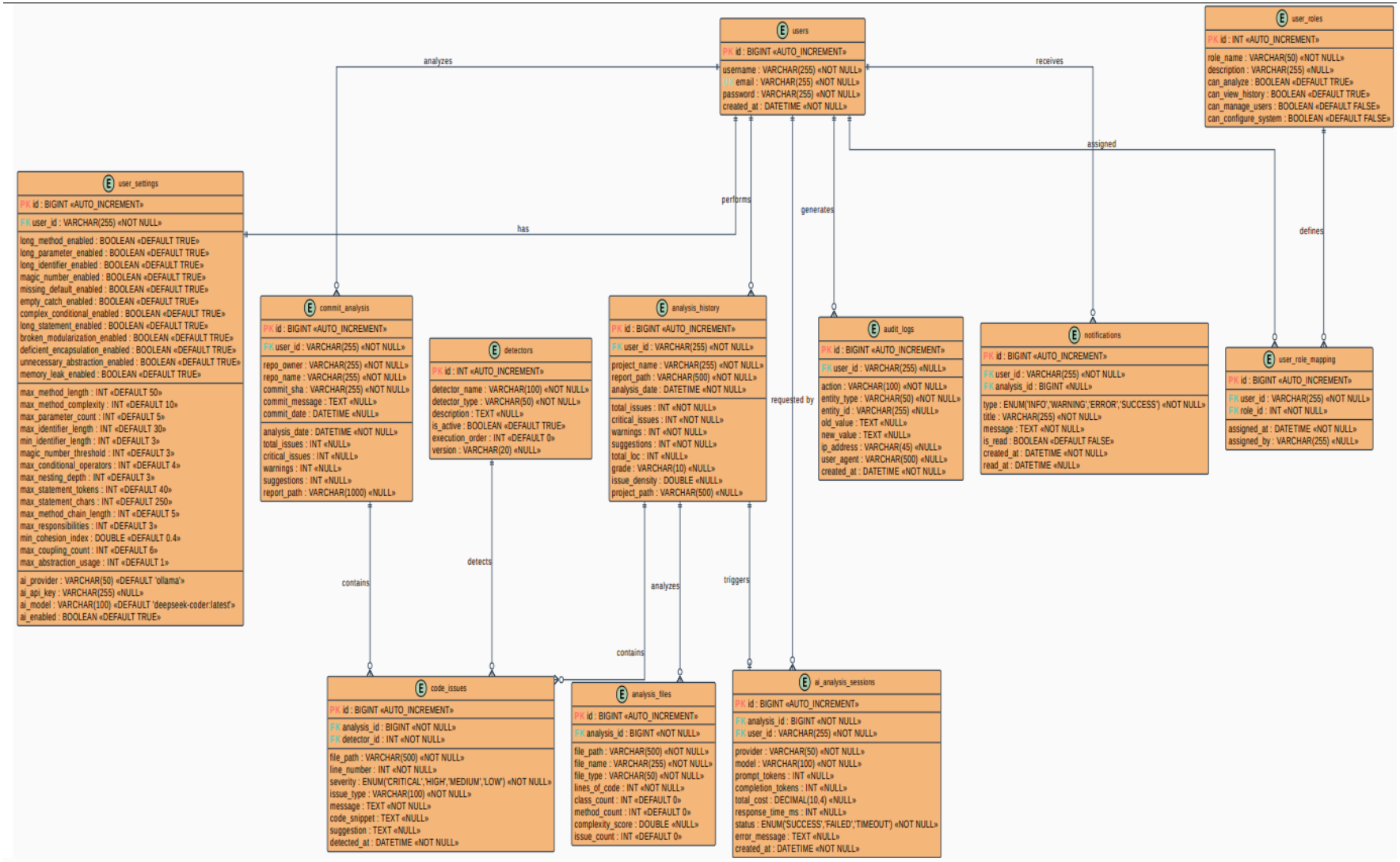

Figure 22: Entity Relationship Diagram

C.2 Sample Queries

C.2.1 Retrieve User Analysis History

```
SELECT * FROM analysis_history
WHERE user_id = ?
ORDER BY analysis_date DESC
LIMIT 10;
```

C.2.2 Calculate Monthly Analysis Count

```
SELECT DATE_FORMAT(analysis_date, '%Y-%m') as month,
       COUNT(*) as analyses
FROM analysis_history
GROUP BY month
ORDER BY month DESC;
```

C.2.3 Get Issue Distribution

```
SELECT SUM(critical_issues) as critical,
       SUM(warnings) as warnings,
       SUM(suggestions) as suggestions
FROM analysis_history;
```

D API Endpoints Reference

This appendix documents all REST API endpoints available in the DevSync platform.

D.1 Authentication Endpoints

Table 49: Authentication API Endpoints

Endpoint	Method	Description
/api/auth/signup	POST	Register new user account
/api/auth/login	POST	Authenticate user and return token
/api/auth/users	GET	Get all users (admin only)
/api/auth/profile/:userId	GET	Get user profile information

D.1.1 POST /api/auth/signup

Request Body:

```
{
  "username": "string",
  "email": "string",
  "password": "string"
}
```

Response (200 OK):

```
"User registered successfully"
```

D.1.2 POST /api/auth/login

Request Body:

```
{
  "email": "string",
  "password": "string"
}
```

Response (200 OK):

```
{
  "message": "Login successful",
  "userId": "string",
  "username": "string",
  "token": "string"
}
```

D.2 Code Analysis Endpoints

Table 50: Code Analysis API Endpoints

Endpoint	Method	Description
/api/upload	POST	Upload and analyze project
/api/upload/report	GET	Retrieve analysis report
/api/upload/history	GET	Get user's analysis history
/api/upload/visual	POST	Generate visual report
/api/upload/fix-counts	POST	Fix report issue counts

D.2.1 POST /api/upload

Request (multipart/form-data):

```
file: [ZIP/JAR file]
userId: "string"
```

Response (200 OK):

```
" Advanced Analysis Complete!
 Extracted to: uploads/[folder]
 Java files: [count]
 Lines of Code: [count]
 Report: [filename]
 Issues detected: [count]
 Grade: [grade] ([score]%)
 Issue Density: [density] issues/KLOC
 Quality: [level]
 AI analysis: [status]
 Report path: [path]"
```

D.2.2 GET /api/upload/report

Query Parameters:

```
path: "string" (report file path)
userId: "string"
```

Response (200 OK):

```
[Report content as text]
```

D.3 GitHub Integration Endpoints

Table 51: GitHub API Endpoints

Endpoint	Method	Description
/api/github/test	GET	Test GitHub API connectivity
/api/github/repos	POST	Get user's repositories
/api/github/commits	POST	Get repository commits
/api/github/download	POST	Download repository
/api/github/analyze-commit	POST	Analyze specific commit
/api/github/commit-history	GET	Get commit analysis history

D.3.1 POST /api/github/repos

Request Body:

```
{
  "token": "string"
}
```

Response (200 OK):

```
[
  {
    "name": "string",
    "full_name": "string",
    "owner": { ... },
    "updated_at": "timestamp",
    ...
  }
]
```

D.3.2 POST /api/github/analyze-commit

Request Body:

```
{
  "token": "string",
  "owner": "string",
  "repo": "string",
  "sha": "string",
```

```
  "userId": "string",
  "commitMessage": "string",
  "commitDate": "timestamp"
}
```

Response (200 OK):

```
{
  "id": "number",
  "userId": "string",
  "repoOwner": "string",
  "repoName": "string",
  "commitSha": "string",
  "totalIssues": "number",
  "criticalIssues": "number",
  "warnings": "number",
  "suggestions": "number",
  "reportPath": "string",
  "analysisDate": "timestamp"
}
```

D.4 Admin Endpoints

Table 52: Admin API Endpoints

Endpoint	Method	Description
/api/admin/dashboard	GET	Get dashboard statistics
/api/admin/users	GET	Get all users
/api/admin/users/:userId	GET	Get user details
/api/admin/users/:userId	PUT	Update user
/api/admin/users/:userId	DELETE	Delete user
/api/admin/projects	GET	Get all projects
/api/admin/reports	GET	Get all reports
/api/admin/settings	GET	Get admin settings
/api/admin/settings	POST	Save admin settings
/api/admin/settings/init	POST	Initialize default settings
/api/admin/fix-counts	POST	Fix report counts

D.4.1 GET /api/admin/dashboard

Response (200 OK):

```
{
  "totalUsers": "number",
  "totalIssues": "number",
  "aiAnalysisCount": "number",
  "criticalIssues": "number",
  "warningIssues": "number",
  "suggestionIssues": "number",
  "cleanFiles": "number"
}
```

D.5 Settings Endpoints

Table 53: Settings API Endpoints

Endpoint	Method	Description
/api/settings/:userId	GET	Get user settings
/api/settings/:userId	POST	Save user settings

D.6 Error Responses

Table 54: Common Error Responses

Status Code	Meaning	Example Response
400	Bad Request	"Invalid input data"
401	Unauthorized	"Authentication required"
403	Forbidden	"Access denied"
404	Not Found	"Resource not found"
500	Internal Server Error	"Server error occurred"
503	Service Unavailable	"System under maintenance"

D.7 CORS Configuration

All endpoints support CORS with the following configuration:

```
Allowed Origins: http://localhost:5173, * (configurable)
Allowed Methods: GET, POST, PUT, DELETE, OPTIONS
Allowed Headers: Content-Type, Authorization
```

E Code Smell Detection Rules

This appendix provides detailed detection rules and thresholds for all twelve code smell detectors.

E.1 Long Method Detector

E.1.1 Detection Criteria

Table 55: Long Method Detection Thresholds

Metric	Threshold	Severity
Lines of Code (LOC)	> 100	Critical
Cyclomatic Complexity	> 10	High
Cognitive Complexity	> 15	High
Nesting Depth	> 4	Medium

E.1.2 Calculation Formulas

Cyclomatic Complexity:

```
complexity = 1 + count(if) + count(for) + count(while)
           + count(case) + count(&&) + count(||)
```

Cognitive Complexity:

```
cognitive = sum(nesting_penalties) + sum(structural_penalties)
where nesting_penalty = nesting_level for each control structure
```

E.2 Magic Number Detector

E.2.1 Detection Rules

Table 56: Magic Number Detection Rules

Rule	Description
Hardcoded literals	Numeric literals not in constants
Allowed values	0, 1, -1 (common programming idioms)
Context exceptions	Array indices, loop counters

E.3 Empty Catch Block Detector

E.3.1 Detection Criteria

- Catch block with no statements
- Catch block with only comments
- Catch block with only TODO markers

E.4 Broken Modularization Detector

E.4.1 Cohesion Metrics

Table 57: Cohesion Thresholds

Metric	Threshold	Interpretation
Cohesion Index	< 0.4	Low cohesion (God Class)
Cohesion Index	0.4 - 0.6	Moderate cohesion
Cohesion Index	> 0.6	High cohesion (Good)

Cohesion Formula:

```
cohesionIndex = usedFields / totalFields
where usedFields = unique fields accessed by methods
```

E.4.2 Coupling Metrics

Table 58: Coupling Thresholds

Dependencies	Level	Action
< 5	Low coupling	Acceptable
5 - 7	Moderate coupling	Review
> 7	High coupling	Refactor

E.5 Complex Conditional Detector

E.5.1 Detection Rules

Table 59: Complex Conditional Thresholds

Condition	Threshold
Logical operators in single condition	> 3
Nested if statements	> 3 levels
Ternary operator complexity	> 2 nested

E.6 Deficient Encapsulation Detector

E.6.1 Detection Rules

- Public non-final fields
- Protected fields (except in inheritance hierarchies)
- Package-private fields without justification
- Exceptions: public static final constants

E.7 Long Parameter List Detector

E.7.1 Thresholds

Table 60: Parameter List Thresholds

Parameter Count	Severity	Recommendation
<= 3	None	Acceptable
4 - 6	Low	Consider parameter object
7 - 9	Medium	Refactor recommended
>= 10	High	Refactor required

E.8 Long Statement Detector

E.8.1 Detection Criteria

Table 61: Statement Length Thresholds

Characters	Action
<= 120	Acceptable
121 - 150	Warning
> 150	Critical

E.9 Long Identifier Detector

E.9.1 Naming Thresholds

Table 62: Identifier Length Thresholds

Identifier Type	Max Length	Recommendation
Variable	30 characters	Use abbreviations
Method	40 characters	Simplify name
Class	35 characters	Reconsider design

E.10 Missing Default Detector

E.10.1 Detection Rules

- Switch statements without default case
- Enum switches missing cases
- Exception: Complete enum coverage

E.11 Unnecessary Abstraction Detector

E.11.1 Detection Criteria

Table 63: Abstraction Detection Rules

Pattern	Issue
Interface with 1 implementation	Unnecessary abstraction
Abstract class with no subclasses	Unused abstraction
Single-method interface	Consider functional interface

E.12 Memory Leak Detector

E.12.1 Detection Patterns

Table 64: Memory Leak Patterns

Pattern	Description
Unclosed resources	Streams, connections not closed
Static collections	Growing static lists/maps
Event listeners	Registered but not removed
Thread locals	Not cleaned up
Inner class references	Implicit outer class reference

E.13 Severity Classification

Table 65: Issue Severity Levels

Severity	Description
Critical	Severe issues requiring immediate attention (security, memory leaks, major design flaws)
High	Important issues affecting maintainability (complex methods, high coupling)
Medium	Moderate issues affecting code quality (long parameters, missing defaults)
Low	Minor issues and suggestions (naming conventions, minor improvements)

E.14 Grading Scale

Table 66: Code Quality Grading Scale

Grade	Score	Issue Density	Quality Level
A+	95-100	< 0.5 issues/KLOC	Excellent
A	90-94	0.5-1.0 issues/KLOC	Very Good
B	80-89	1.0-2.0 issues/KLOC	Good
C	70-79	2.0-5.0 issues/KLOC	Acceptable
D	60-69	5.0-10.0 issues/KLOC	Below Average
F	0-59	> 10.0 issues/KLOC	Poor

F Installation and Deployment Guide

This appendix provides step-by-step instructions for installing, configuring, and deploying the DevSync platform.

F.1 Prerequisites

F.1.1 Required Software

Table 67: Required Software Components

Software	Version	Purpose
JDK	21 or higher	Java runtime environment
Node.js	18.x or higher	Frontend build tools
MySQL	8.0 or higher	Database server
Maven	3.8 or higher	Backend build tool
Git	Latest	Version control

F.1.2 Optional Software

Table 68: Optional Software Components

Software	Version	Purpose
Ollama	Latest	AI-powered analysis
Docker	Latest	Containerized deployment

F.2 Backend Installation

F.2.1 Step 1: Clone Repository

```
git clone https://github.com/your-repo/devsync.git
cd devsync
```

F.2.2 Step 2: Configure Database

Create MySQL database:

```
mysql -u root -p
CREATE DATABASE devsyncdb;
CREATE USER 'devsync'@'localhost' IDENTIFIED BY 'password';
GRANT ALL PRIVILEGES ON devsyncdb.* TO 'devsync'@'localhost';
FLUSH PRIVILEGES;
EXIT;
```

F.2.3 Step 3: Configure Application Properties

Edit `src/main/resources/application.properties`:

```
# Database Configuration
spring.datasource.url=jdbc:mysql://localhost:3306/devsyncdb
spring.datasource.username=devsync
spring.datasource.password=password
spring.jpa.hibernate.ddl-auto=update

# Server Configuration
server.port=8080

# File Upload Configuration
spring.servlet.multipart.max-file-size=50MB
spring.servlet.multipart.max-request-size=50MB

# Security Configuration
spring.security.user.name=admin
spring.security.user.password=admin123
```

F.2.4 Step 4: Build Backend

```
mvn clean install
```

F.2.5 Step 5: Run Backend

```
mvn spring-boot:run
```

Or run the JAR file:

```
java -jar target/devsync-0.0.1-SNAPSHOT.jar
```

F.3 Frontend Installation

F.3.1 Step 1: Navigate to Frontend Directory

```
cd frontend
```

F.3.2 Step 2: Install Dependencies

```
npm install
```

F.3.3 Step 3: Configure Environment

Create `.env` file:

```
VITE_API_URL=http://localhost:8080
VITE_APP_NAME=DevSync
```

F.3.4 Step 4: Run Development Server

```
npm run dev
```

Access at: `http://localhost:5173`

F.3.5 Step 5: Build for Production

```
npm run build
```

Production files will be in `dist/` directory.

F.4 Ollama AI Setup (Optional)

F.4.1 Step 1: Install Ollama

Download from: `https://ollama.ai`

F.4.2 Step 2: Pull AI Model

```
ollama pull deepseek-coder:latest
```

F.4.3 Step 3: Start Ollama Service

```
ollama serve
```

Service runs on: `http://localhost:11434`

F.5 Production Deployment

F.5.1 Backend Deployment

Option 1: Standalone JAR

```
java -jar devsync.jar --spring.profiles.active=prod
```

Option 2: Docker Container

Create `Dockerfile`:

```
FROM openjdk:21-jdk-slim
WORKDIR /app
COPY target/devsync-0.0.1-SNAPSHOT.jar app.jar
EXPOSE 8080
ENTRYPOINT ["java", "-jar", "app.jar"]
```

Build and run:

```
docker build -t devsync-backend .
docker run -p 8080:8080 devsync-backend
```

F.5.2 Frontend Deployment

Option 1: Static Hosting

Deploy `dist/` folder to:

- Netlify
- Vercel
- AWS S3 + CloudFront
- GitHub Pages

Option 2: Nginx Server

Nginx configuration:

```
server {
    listen 80;
    server_name devsync.example.com;
    root /var/www/devsync/dist;
    index index.html;
```

```
    location / {
        try_files $uri $uri/ /index.html;
    }

    location /api {
        proxy_pass http://localhost:8080;
    }
}
```

F.6 Troubleshooting

F.6.1 Common Issues

Table 69: Common Issues and Solutions

Issue	Solution
Database connection failed	Verify MySQL is running, check credentials in application.properties
Port 8080 already in use	Change server.port in application.properties or stop conflicting service
CORS errors	Verify CORS configuration in backend allows frontend origin
File upload fails	Check max-file-size configuration and disk space
Ollama not responding	Ensure Ollama service is running on port 11434
Frontend build fails	Clear node_modules and reinstall: npm clean-install

G User Manual

This appendix provides end-user instructions for using the DevSync platform.

G.1 Getting Started

G.1.1 Creating an Account

1. Navigate to the DevSync landing page
2. Click "Sign Up" or "Get Started"

3. Enter username, email, and password
4. Click "Register" to create account
5. You will be redirected to the login page

G.1.2 Logging In

1. Enter your email and password
2. Click "Login"
3. You will be redirected to the dashboard

G.2 Analyzing Code

G.2.1 Uploading a Project

1. From the dashboard, locate the upload area
2. Either:
    - Drag and drop a ZIP file into the upload zone, or
    - Click "Browse" to select a file from your computer
3. Ensure the file is a ZIP or JAR file containing Java source code
4. File size must not exceed 50MB
5. Click "Analyze Project" to start the analysis

G.2.2 Viewing Analysis Results

1. Wait for the analysis to complete (progress bar will show status)
2. View the summary metrics displayed in cards:
    - Critical Issues (red)
    - Warnings (yellow)
    - Suggestions (blue)
3. Review the overall grade and quality score
4. Click "View Full Report" to see detailed analysis

G.2.3 Understanding the Report

The analysis report contains:

- **Executive Summary:** Overall project statistics and grade
- **Metrics Summary:** Lines of code, classes, methods, complexity
- **Issue Breakdown:** Detailed list of detected code smells by file
- **AI Analysis:** Intelligent recommendations for improvement
- **Recommendations:** Actionable steps to improve code quality

G.2.4 Downloading Reports

1. Click the "Download Report" button
2. Choose format (TXT or PDF)
3. Report will be downloaded to your default downloads folder

G.3 Managing Analysis History

G.3.1 Viewing Past Analyses

1. Click "History" in the navigation menu
2. Browse the list of previous analyses
3. Each entry shows:
    - Project name
    - Analysis date and time
    - Issue counts
    - Grade received

G.3.2 Comparing Analyses

1. Select two or more analyses from history
2. Click "Compare" button
3. View side-by-side comparison showing:
    - Issue count trends
    - Grade improvements
    - Code quality metrics over time

G.4 GitHub Integration

G.4.1 Connecting GitHub Account

1. Navigate to Settings
2. Click "Connect GitHub"
3. Enter your GitHub Personal Access Token
4. Click "Connect"
5. Your repositories will be loaded

G.4.2 Analyzing a Repository

1. From the GitHub tab, select a repository
2. View the list of commits
3. Click "Analyze" next to any commit
4. Wait for analysis to complete
5. View results in the commit history section

G.5 Customizing Settings

G.5.1 Analysis Settings

1. Click "Settings" in the navigation menu
2. Configure detection thresholds:
    - Maximum method length
    - Complexity thresholds
    - Parameter count limits
3. Enable/disable specific detectors
4. Click "Save Settings"

G.5.2 AI Settings

1. Navigate to Settings > AI Configuration
2. Toggle AI analysis on/off
3. Select AI provider (Ollama, OpenAI, etc.)
4. Configure AI model preferences
5. Save changes

G.5.3 Theme Settings

1. Click the theme toggle button in the header
2. Choose between:
    - Light mode
    - Dark mode
    - System preference
3. Theme preference is saved automatically

G.6 Tips for Best Results

- **Project Structure:** Ensure your ZIP file contains a standard Java project structure
- **File Size:** Keep projects under 50MB for optimal performance
- **Regular Analysis:** Analyze code frequently to track improvements
- **Review AI Suggestions:** AI recommendations provide valuable insights
- **Compare Over Time:** Use history comparison to measure progress
- **Customize Thresholds:** Adjust detection thresholds to match your team's standards

H Future Enhancements

This appendix outlines potential future enhancements and features for the DevSync platform.

H.1 Planned Features

H.1.1 Multi-Language Support

- Extend analysis capabilities to Python, JavaScript, C++, and C#
- Language-specific detector implementations
- Unified reporting across multiple languages
- Cross-language project analysis

H.1.2 Real-Time Collaboration

- Team workspaces for shared projects
- Real-time analysis result sharing
- Collaborative code review features
- Team performance dashboards
- Role-based access control for team members

H.1.3 IDE Integration

- IntelliJ IDEA plugin
- Visual Studio Code extension
- Eclipse plugin
- Real-time code analysis as you type
- Inline suggestions and quick fixes

H.1.4 CI/CD Integration

- Jenkins plugin
- GitHub Actions integration
- GitLab CI/CD pipeline support
- Automated quality gates
- Pull request analysis and comments

H.1.5 Advanced Analytics

- Technical debt calculation and tracking
- Code quality trends over time
- Predictive analysis for potential issues
- Team productivity metrics
- Custom reporting and dashboards

H.1.6 Enhanced AI Capabilities

- Automated code refactoring suggestions
- AI-powered code generation for fixes
- Natural language query interface
- Context-aware learning from user feedback
- Integration with multiple AI providers (GPT-4, Claude, etc.)

H.1.7 Security Enhancements

- Advanced vulnerability scanning
- OWASP Top 10 detection
- Dependency vulnerability analysis
- Security best practices enforcement
- Compliance reporting (GDPR, HIPAA, etc.)

H.2 Performance Improvements

- Parallel analysis processing for large projects
- Incremental analysis (analyze only changed files)
- Caching mechanisms for repeated analyses
- Distributed processing for enterprise deployments
- Optimized memory usage for large codebases

H.3 User Experience Enhancements

- Interactive code visualization
- Dependency graph visualization
- Code smell heatmaps
- Guided tutorials and onboarding
- Mobile application for iOS and Android

I Conclusion

DevSync represents a comprehensive solution for automated Java code quality analysis, successfully combining traditional static analysis techniques with modern AI-powered recommendations. The platform addresses the critical need for consistent, automated code review in software development teams.

I.1 Project Achievements

The project successfully delivered:

- **Comprehensive Analysis Engine:** Twelve specialized detectors covering major code smell categories
- **Modern Web Application:** React 18.3.1 + Vite 7.1.7 frontend with Spring Boot backend
- **AI Integration:** Ollama-powered intelligent recommendations with fallback mechanisms
- **GitHub Integration:** Direct repository and commit analysis capabilities
- **User-Friendly Interface:** Intuitive dashboard with dark/light theme support
- **Robust Testing:** 223 test cases with 100% pass rate
- **Complete Documentation:** Comprehensive technical and user documentation

I.2 Technical Contributions

The DevSync platform makes several technical contributions:

1. **Multi-Dimensional Complexity Analysis:** Combines cyclomatic and cognitive complexity metrics for nuanced code quality assessment
2. **Intelligent Grading System:** Issue density-based grading that accounts for project size
3. **Hybrid AI Approach:** Local AI processing with graceful degradation ensures continuous service
4. **Modular Detector Architecture:** Extensible design allows easy addition of new detectors
5. **Modern Frontend Stack:** Leverages latest React ecosystem tools for optimal performance

I.3 Learning Outcomes

Through this project, the development team gained valuable experience in:

- Full-stack web application development
- Abstract Syntax Tree (AST) parsing and analysis
- AI service integration and prompt engineering
- RESTful API design and implementation
- Database design and optimization
- Modern frontend development with React and Vite
- Software testing methodologies
- Project management and iterative development

I.4 Impact and Applications

DevSync can benefit various stakeholders:

- **Individual Developers:** Personal code quality improvement and learning
- **Development Teams:** Consistent code review standards and quality gates
- **Educational Institutions:** Teaching tool for software engineering courses
- **Open Source Projects:** Automated quality checks for contributions
- **Enterprise Organizations:** Code quality monitoring and technical debt management

I.5 Challenges Overcome

The development process involved overcoming several technical challenges:

1. **AST Complexity:** Mastering JavaParser library for accurate code analysis
2. **Performance Optimization:** Efficient analysis of large codebases
3. **AI Integration:** Handling service unavailability and timeout scenarios
4. **Frontend-Backend Coordination:** Seamless data flow and error handling
5. **User Experience:** Balancing feature richness with interface simplicity

I.6 Future Directions

The platform provides a solid foundation for future enhancements including multi-language support, IDE integration, CI/CD pipeline integration, and advanced analytics capabilities. The modular architecture ensures that new features can be added without disrupting existing functionality.

I.7 Final Remarks

DevSync demonstrates the successful application of software engineering principles to create a practical, production-ready code quality analysis platform. The combination of traditional static analysis with AI-powered insights provides developers with actionable recommendations for improving code quality. The comprehensive testing and documentation ensure the platform's reliability and maintainability for future development.

The project successfully meets all specified requirements and provides a valuable tool for the software development community. With its modern architecture, extensible design, and user-friendly interface, DevSync is well-positioned to evolve and adapt to future needs in code quality analysis.